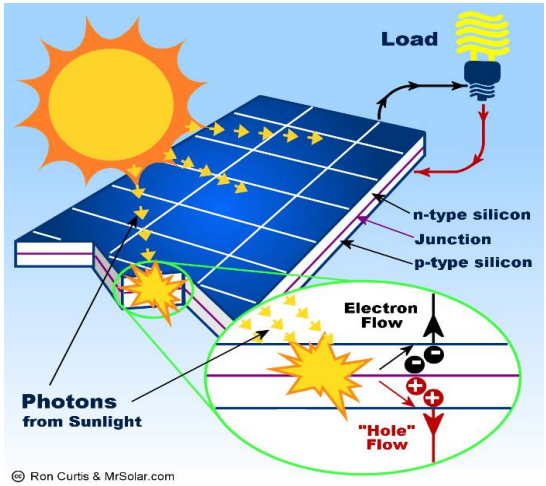
Sustainable Design

By

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India

Possible ways of Solar Energy Conversion



- Solar cells
- 22 % max efficiency

Solar

to

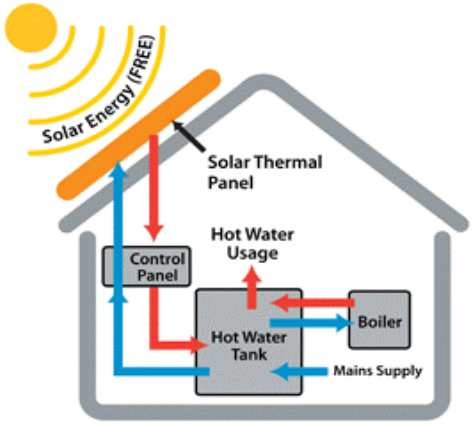
Electricity

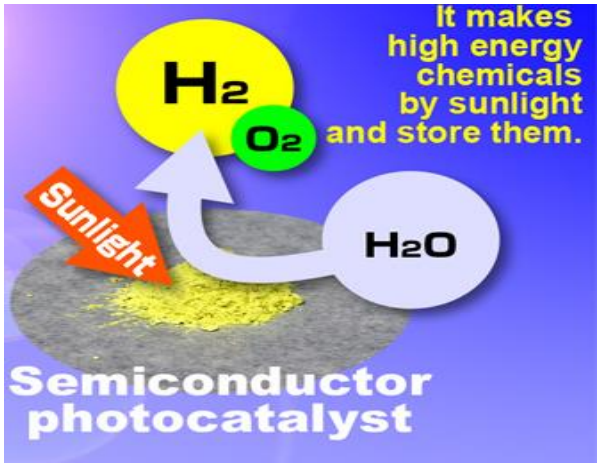
- Solar water heater
- Solar drying
- Solar cooking
- Solar distillation
- Over 30 % max efficiency

Solar

to

Thermal





- Artificial Photosynthesis
- Solar water splitting

Solar

to

Chemical Fuels

Advantages

- It is clean and non-polluting It is a renewable energy
- Solar cells do not produce noise and they are totally silent.
- They require very little maintenance
- They are long lasting sources of energy which can be used almost anywhere
- They have long life time
- There are no fuel costs or fuel supply problems

Disadvantage

Solar power can`t be obtained in night time Solar cells (or) solar panels are very expensive
Energy has not be stored in batteries

They need large area of land to produce more efficient power supply

Solar Cells

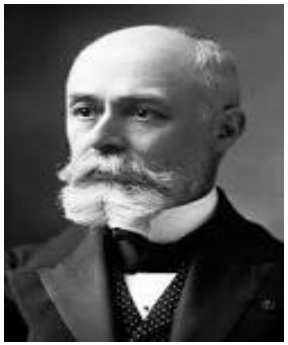
Definition: Solar cell is also called as Photo voltaic cell.

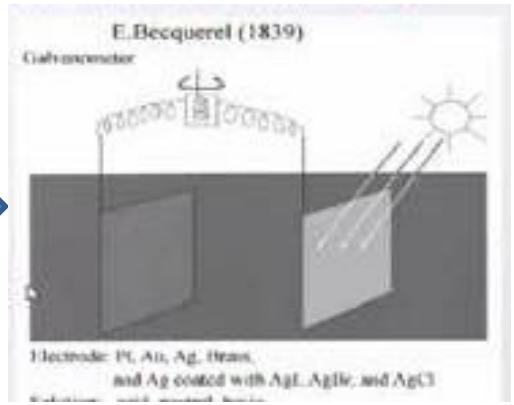
solar cell is the one which converts the solar energy (energy obtained from the sun) directly into electrical energy.

Principle: The solar cells are working based on the principles of photovoltaic effect

In 1839, a French Physicist, Alexandre-Edmund Becquerel

Alexandre Edmond Becquerel (1839)





Photovoltaic effect is the generation of voltage and electric current in a material upon exposure to light

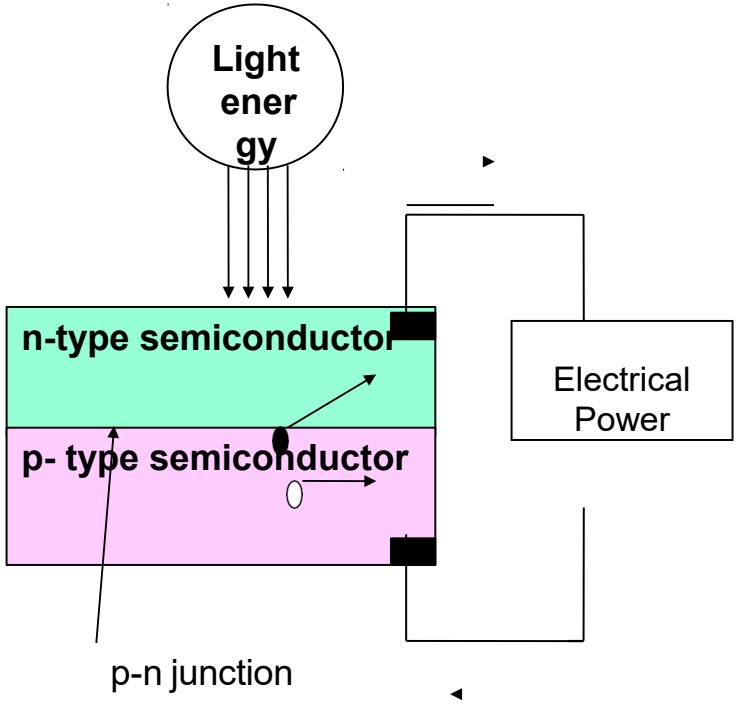
Beginning of Solar Energy Conversion !!!!

Definition:

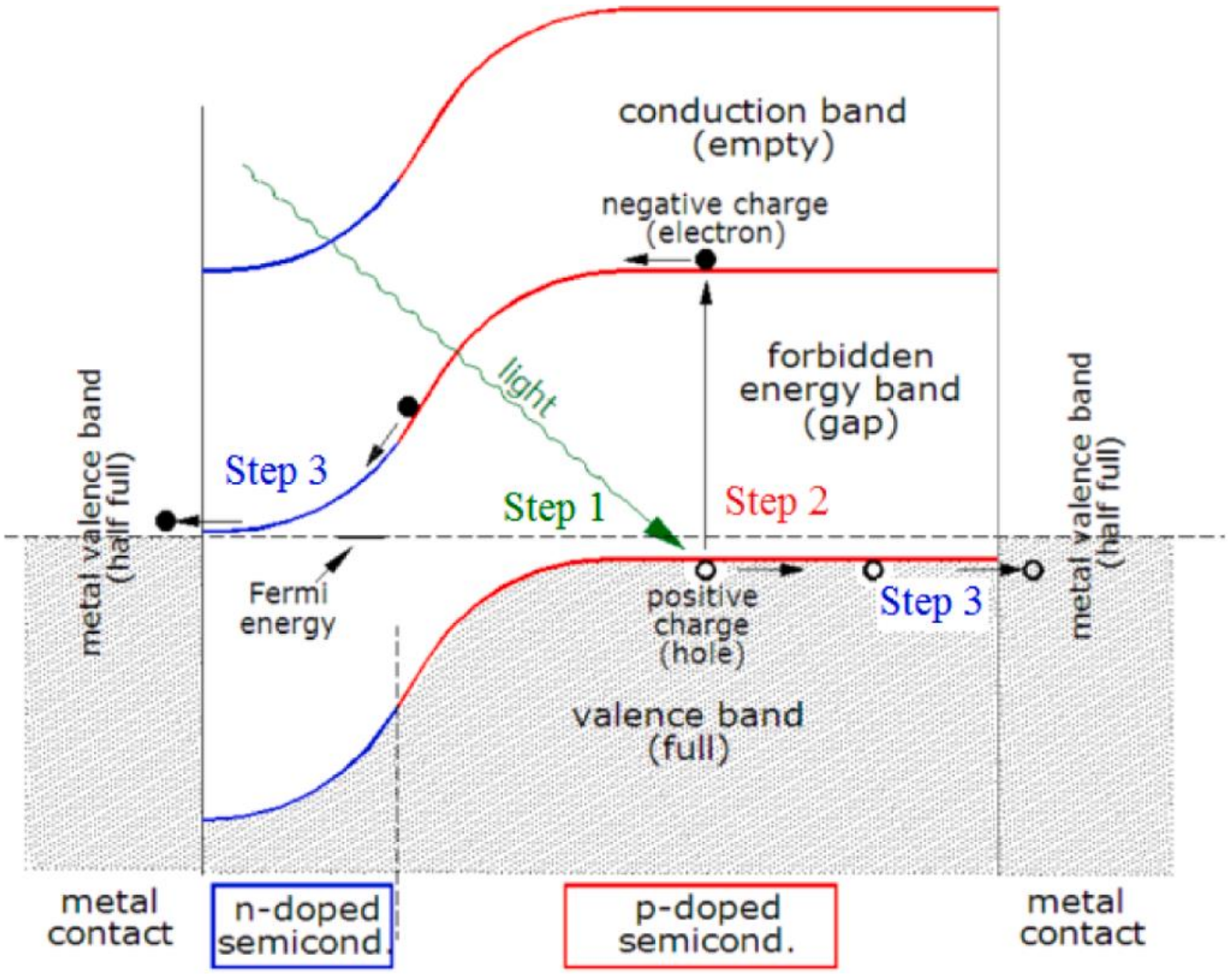
The generation of voltage across the PN junction in a semiconductor due to the absorption of light radiation is called photovoltaic effect.

The Devices based on this effect is called photovoltaic device.

A solar cell operates under no external bias (or in forward bias) to generate power, while a diode is usually used for rectification or switching



The working of a photovoltaic (PV) cell basically involves three steps
: 1 > The absorption of light, creating electron-hole pairs either free or bound (excitons).
2 > The separation of charge carriers.
3 > The separate extraction of these carriers to an external circuit.



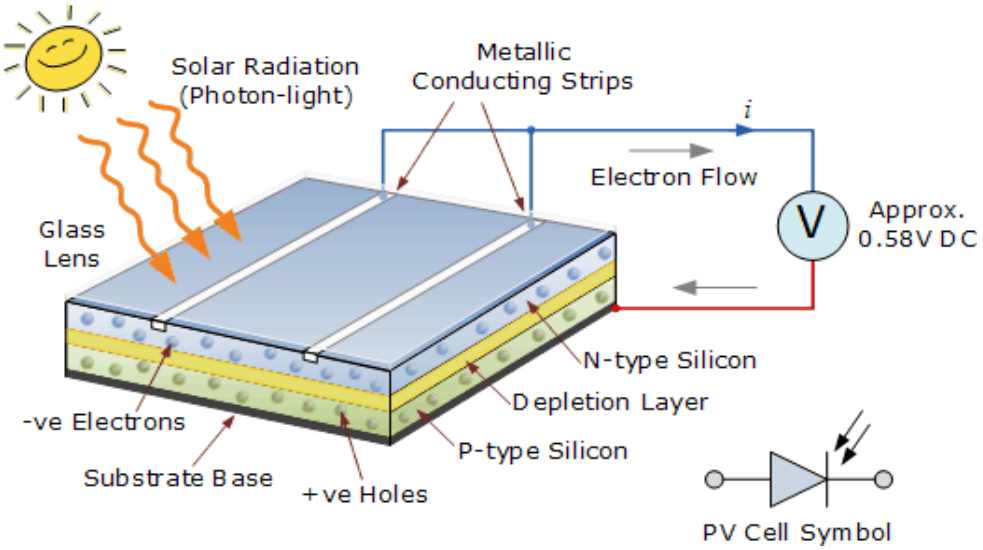
History

PV effect was first studied in solids like Selenium in 1870. It was costly and having less efficiency (1% to 2%) .

In 1940 and in early 1950, Czochralski method was developed for obtaining pure Silicon crystal.

With use of this method crystalline Silicon solar cells were manufactured which were having efficiency up to 11%

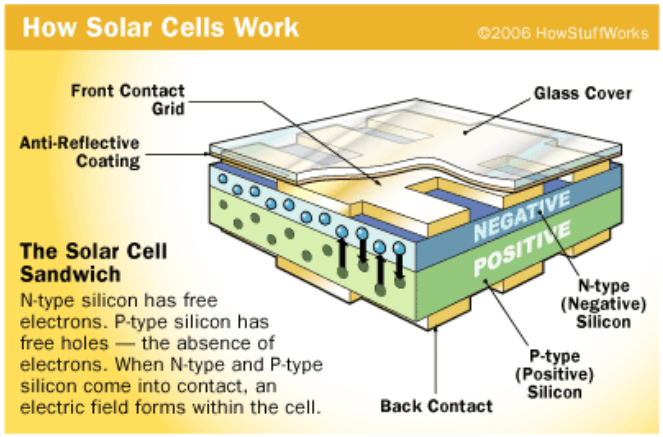
Construction

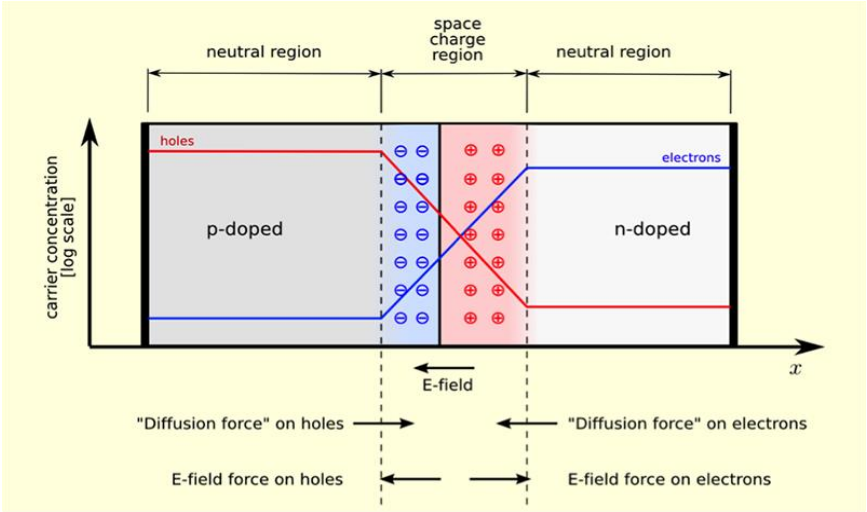


Solar cell (crystalline Silicon) consists of a n-type semiconductor (emitter) layer and p-type semiconductor layer (base).

The two layers are sandwiched and hence there is formation of p-n junction.

The surface is coated with anti-reflection coating to avoid the loss of incident light energy due to reflection.

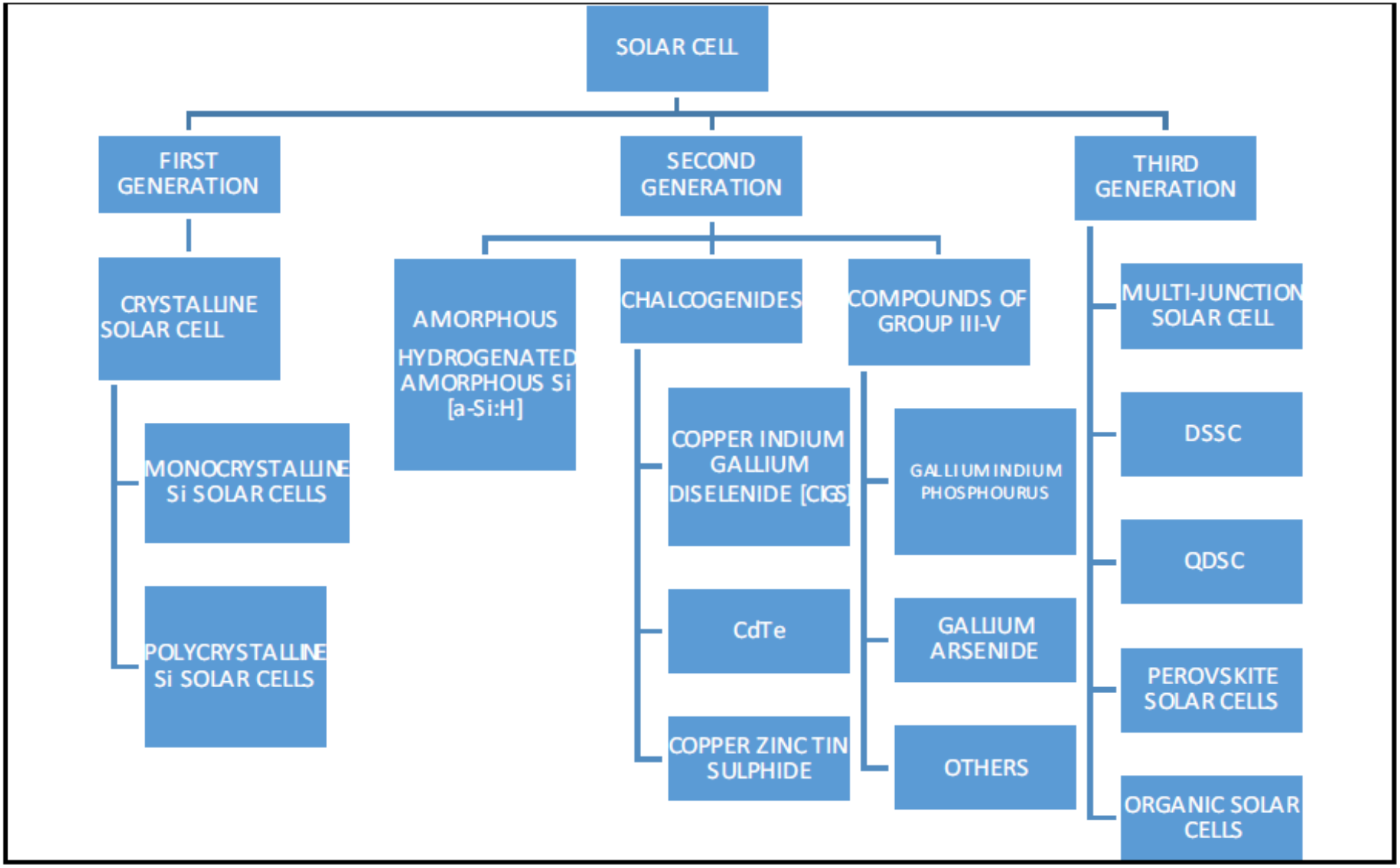




- When a solar panel exposed to sunlight , the light energies are absorbed by a semi conduction materials.
- Due to this absorbed energy, the electrons are liberated and produce the external DC current.
- The DC current is converted into 240-volt AC current using an inverter for different applications.

- Thus when this p and n layers are connected to external circuit, electrons flow from n-layer to p-layer, an hence current is generated
- The electrons that leave the solar cell as current give up their energy to whatever is connected to the solar cell, and then re-enter the solar cell. Once back in the solar cell, the process begins again.

Types of Solar cell



The first generation cells—also called conventional, traditional or wafer-based cells
Made of crystalline silicon,
The commercially predominant PV technology, that includes materials such as polysilicon and monocrystalline silicon.

Second generation cells are thin film solar cells, that include amorphous silicon, CdTe and CIGS cells and are

commercially significant in utility-scale photovoltaic power stations,

building integrated photovoltaics or in small standalone power system.

The third generation of solar cells includes a number of thin-film technologies often described as emerging photovoltaics—most of them have not yet been commercially applied and are still in the research or development phase.

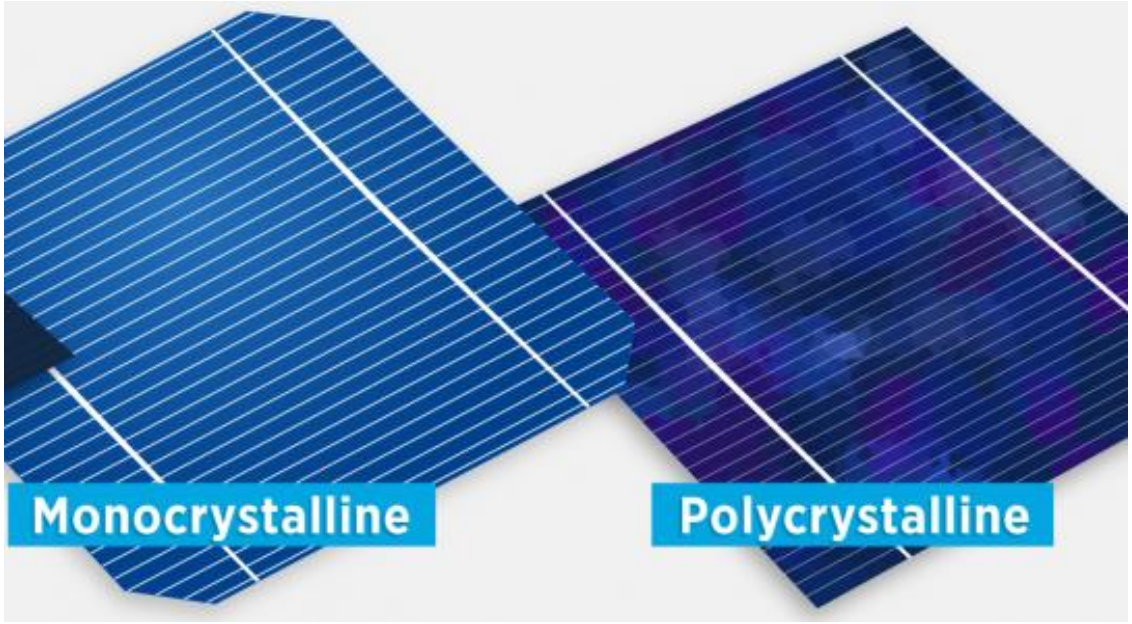
1st generation

Based on the types of crystal used, soar cells can be classified as,

- *Mono crystalline silicon cells*
  - *Semi-crystalline Silicium Solar Cells*
  - *Ribbon Silicium Solar Cells*
- *Polycrystalline silicon cells*

***The Mono crystalline silicon cell** is produced from pure silicon (single crystal).*

Since the Mono crystalline silicon is pure and defect free, the efficiency of cell will be higher.



Semi-crystalline Silicium Solar Cells

This type of solar cells gather from combined small silicium crystallines with cooling of liquid silicium solution. Efficiency of them is near 14% and it is bounded on losses at the edge of combined silicium particles.

Ribbon Silicium Solar Cells

These cells are made of from silicium layers as plate to reduce substance losses. They get with different types of methods and still in improvement phase. Laboratory efficiency changes in between 13 and 14%.

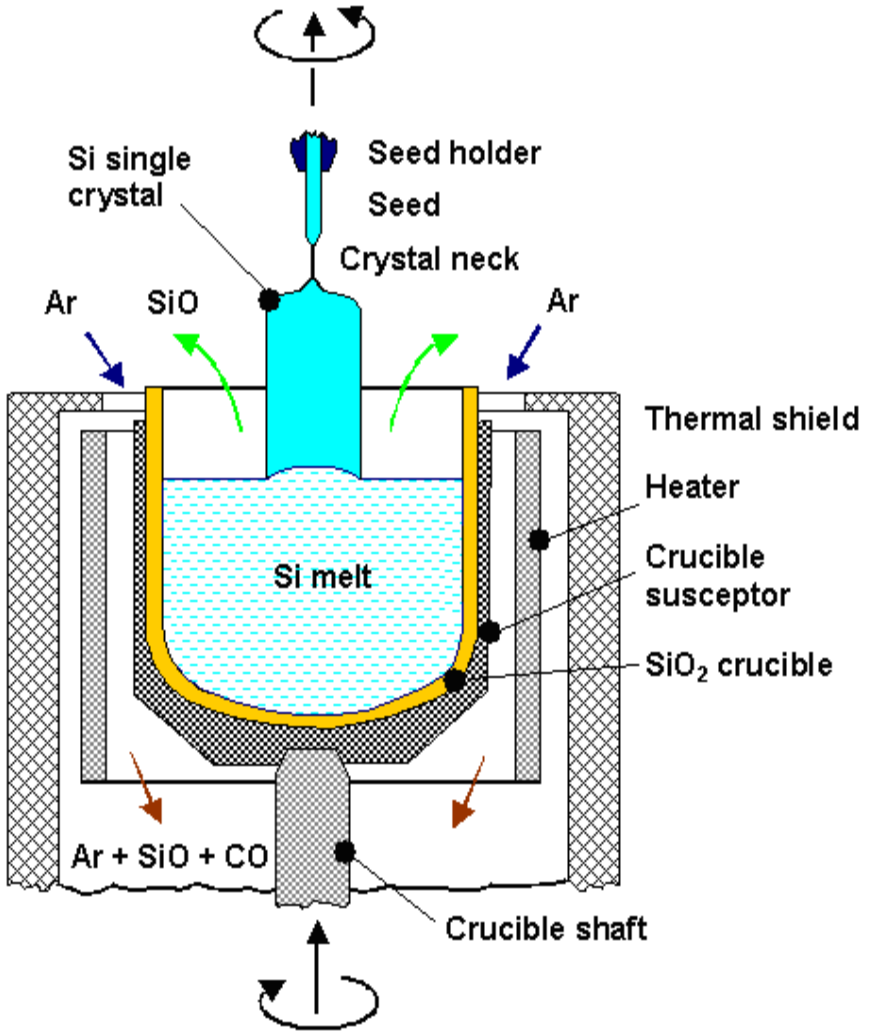
Polycrystalline solar cell

In Polycrystalline solar cell, liquid silicon is used as raw material and polycrystalline silicon was obtained followed by solidification process. The materials contain various crystalline sizes. Hence, the efficiency of this type of cell is less than Mono crystalline cell

Czochralski process is used to manufacture the monocrystalline solar cell

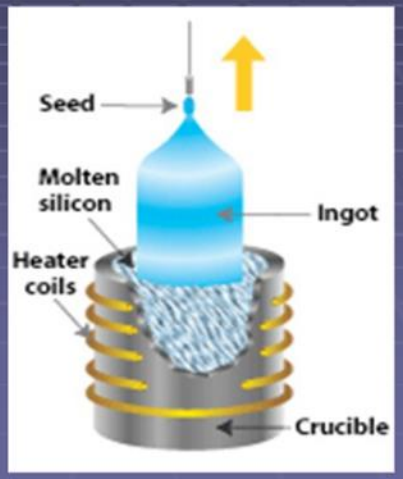
During this Czochralski process, Si crystals are cut from big sized ingots

- High-purity **polycrystalline silicon** is melted in a quartz crucible at ~1420 °C.
- A small single-crystal silicon seed is dipped into the molten silicon.
- The seed crystal is slowly pulled upward while being rotated.
- The molten silicon solidifies onto the seed, replicating its crystal structure.
- Careful control of temperature, pulling rate, and rotation ensures defect-free single crystal growth.
- The grown ingot (cylindrical boule) is typically 1–2 meters long and 200–300 mm in diameter.
- The ingot is cooled and cut into smaller sections.
- These sections are sliced into thin wafers using wire-sawing techniques.
- Wafers are polished and processed further for use in solar cells or integrated circuits.



CZOCHRALSKI PROCESS

- This is the process of creating an ingot.
- A small single silicon rod (seed) is placed in an inert gas at high temps.
- When the seed is rotated up and out silicon adheres to it to form an ingot.

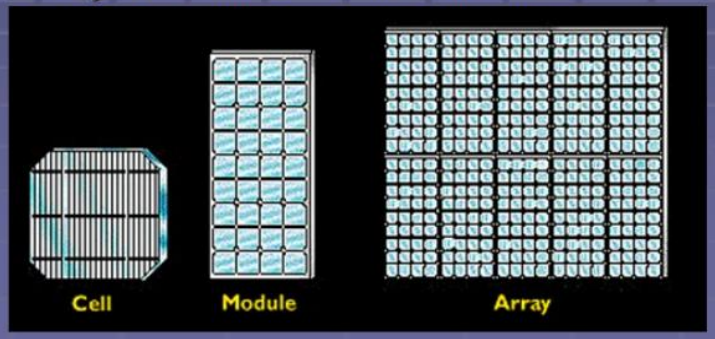


CELLS -> MODULES

Wafers 5 inches square and .012 inches thick are sliced from the ingot.

They are then processed into cells and soldered together to achieve the desired voltage.

Cells arrayed in series are called modules.



TECHNOLOGY

- PV cells exhibit voltage or current when exposed to light.
- The light liberates electrons which move through the cell creating current.
- The larger area there is the more current.
- Single crystal silicon cells are the most efficient at 15-24% sunlight-to-electricity conversion rate.
- They are also the most expensive to produce at $6.50/ watt.

https://www.youtube.com/watch?v=xftnhfa-Dmo

Amorphous Silicon Solar Cell (A-Si)

well developed of the thin film technologies having been on the market for more than 15 years.

It is widely used in pocket calculators, but it also powers some private homes, buildings, and remote facilities

Amorphous Silicon

- The term "amorphous" commonly applied to non-crystalline materials prepared by deposition from gases.
- Non-crystalline:
  - Chemical bonding of atoms nearly unchanged from crystals
  - Small, disorderly variation in the angles between the bonds eliminates regular lattice structure

Hydrogenated Amorphous Silicon

- In early studies of amorphous silicon, it was determined that plasma-deposited amorphous silicon contained a significant percentage of hydrogen atoms bonded into the amorphous silicon structure.
- These atoms were discovered to be essential to the improvement of the electronic properties of the material.
- Amorphous silicon is generally known as "hydrogenated amorphous silicon", or a-Si:H.

ADVANTAGES OF A-SI:H OVER C-SI

- Technology is relatively simple and inexpensive for a-Si:H
- For a given layer thickness, a-Si:H absorbs much more energy than c-Si (about 2.5 times)
- Much less material required for a-Si:H films, lighter weight and less expensive
- Can be deposited on a wide range of substrates, including flexible, curved, and roll-away types
- Overall efficiency of around 10%, still lower than crystalline silicon but improving

Single Layer Amorphous solar cells

Amorphous silicon panels are formed by vapor-depositing a thin layer of silicon material – about 1 micrometer thick – on a substrate material such as glass or metal.

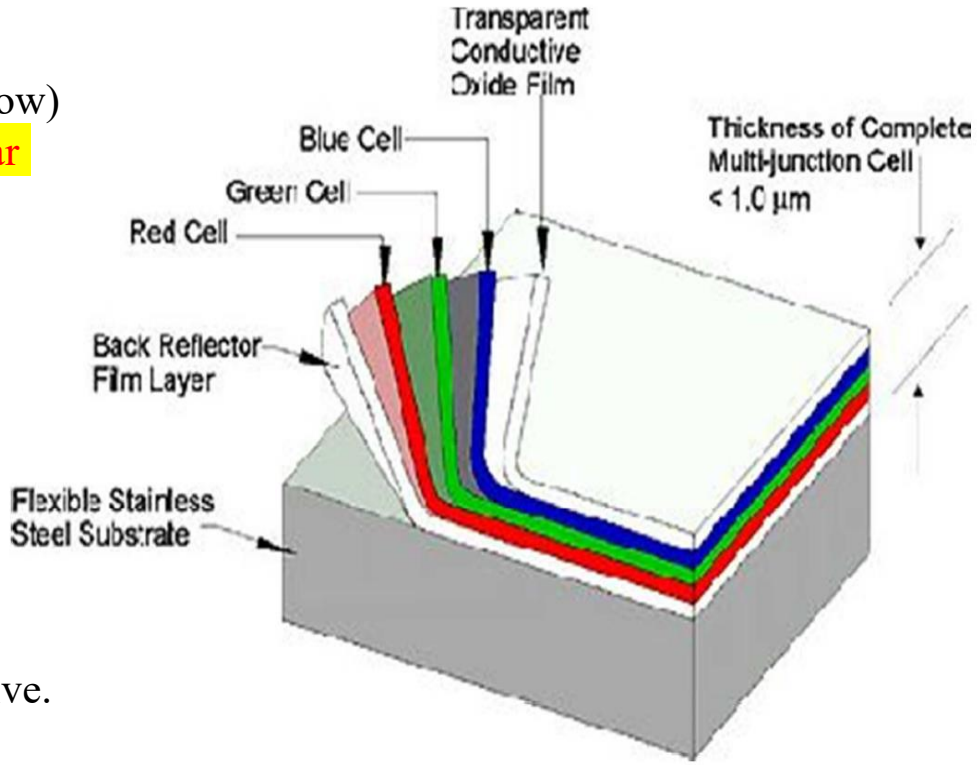
Amorphous silicon can also be deposited at very low temperatures, as low as 75 degrees Celsius, which allows for deposition on plastic as well.

In its simplest form, the cell structure has a single sequence of p-i-n layers.

However, single layer cells suffer from significant degradation in their power output (in the range 15-35%) when exposed to the sun.

Triple layer system

They use a triple layer system (see illustration below) that is optimized to capture light from the full solar spectrum



While crystalline silicon achieves a yield of about 18 percent, amorphous solar cells' yield remains at around 7 percent

The principal advantage of amorphous silicon solar cells is their lower manufacturing costs, which makes these cells very cost competitive.

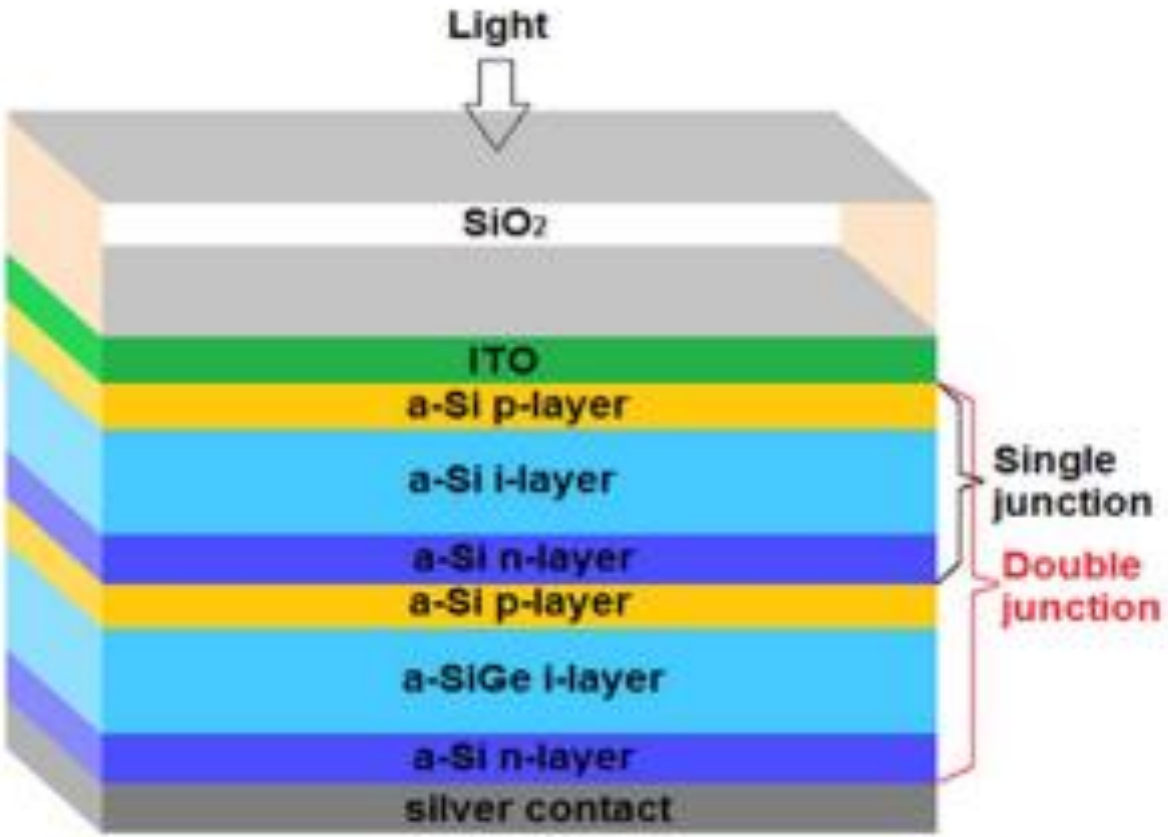
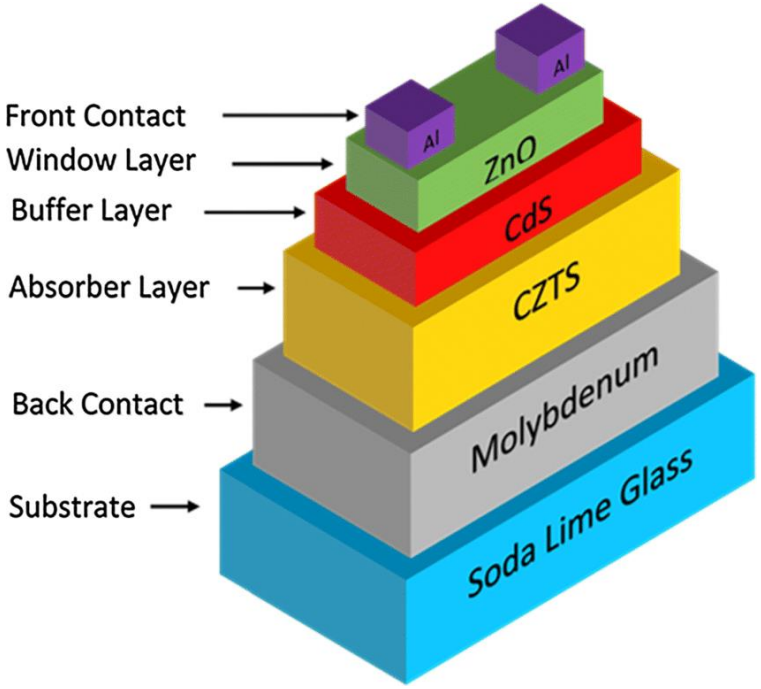
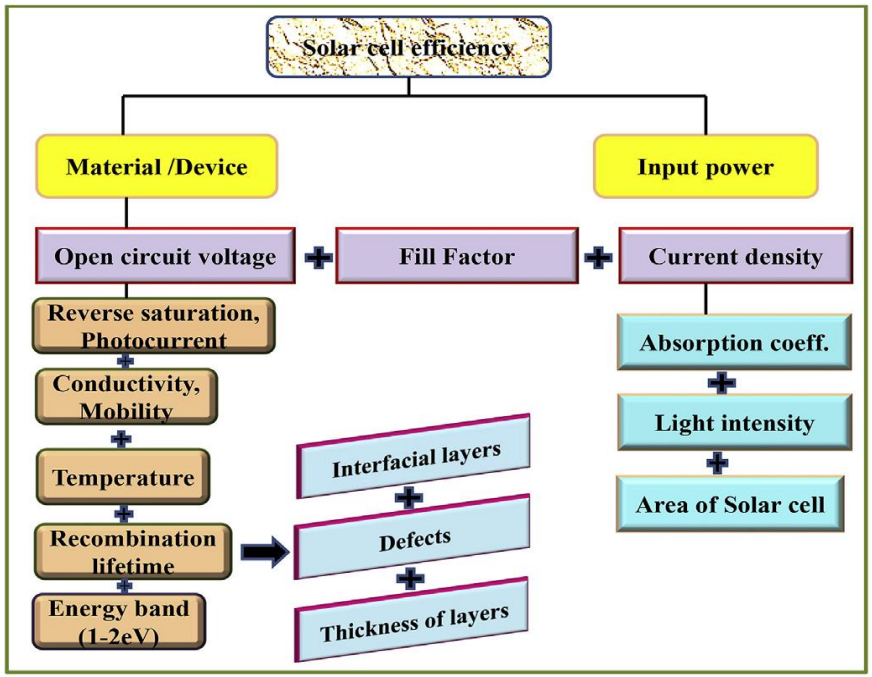


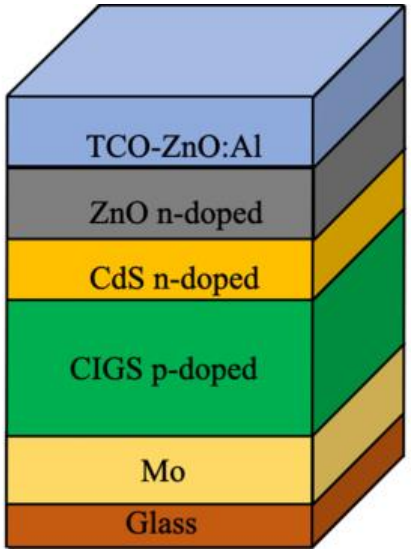
Fig. 4. A layout of a double junction a-Si:H/a-SiGe:H solar cell.

Cu_2ZnSnS_4 (CZTS) thin film for solar cell

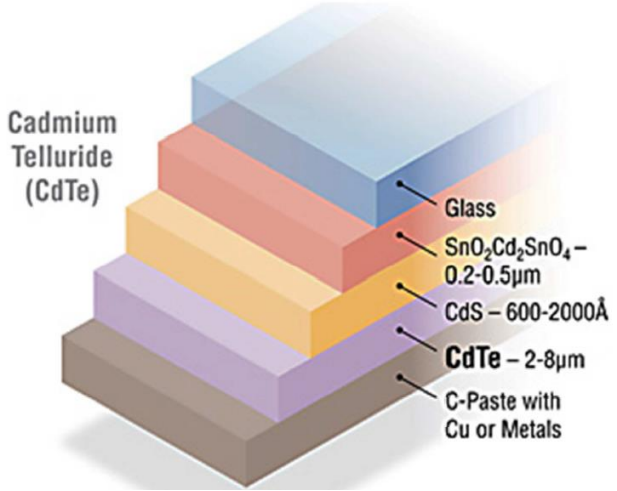




2.5. Toxicity of Cadmium







[CIGS Copper indium gallium selenide solar cell]

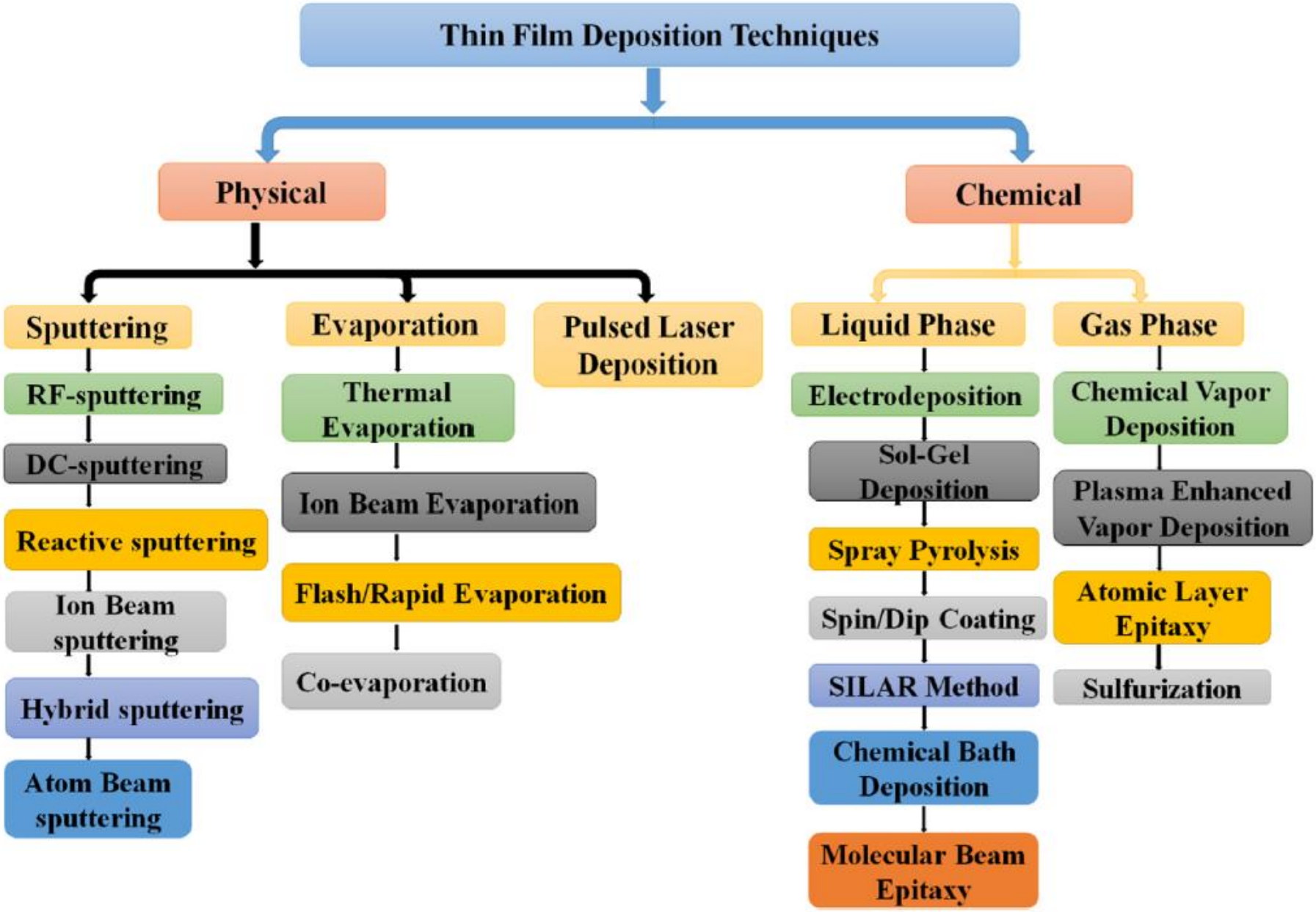
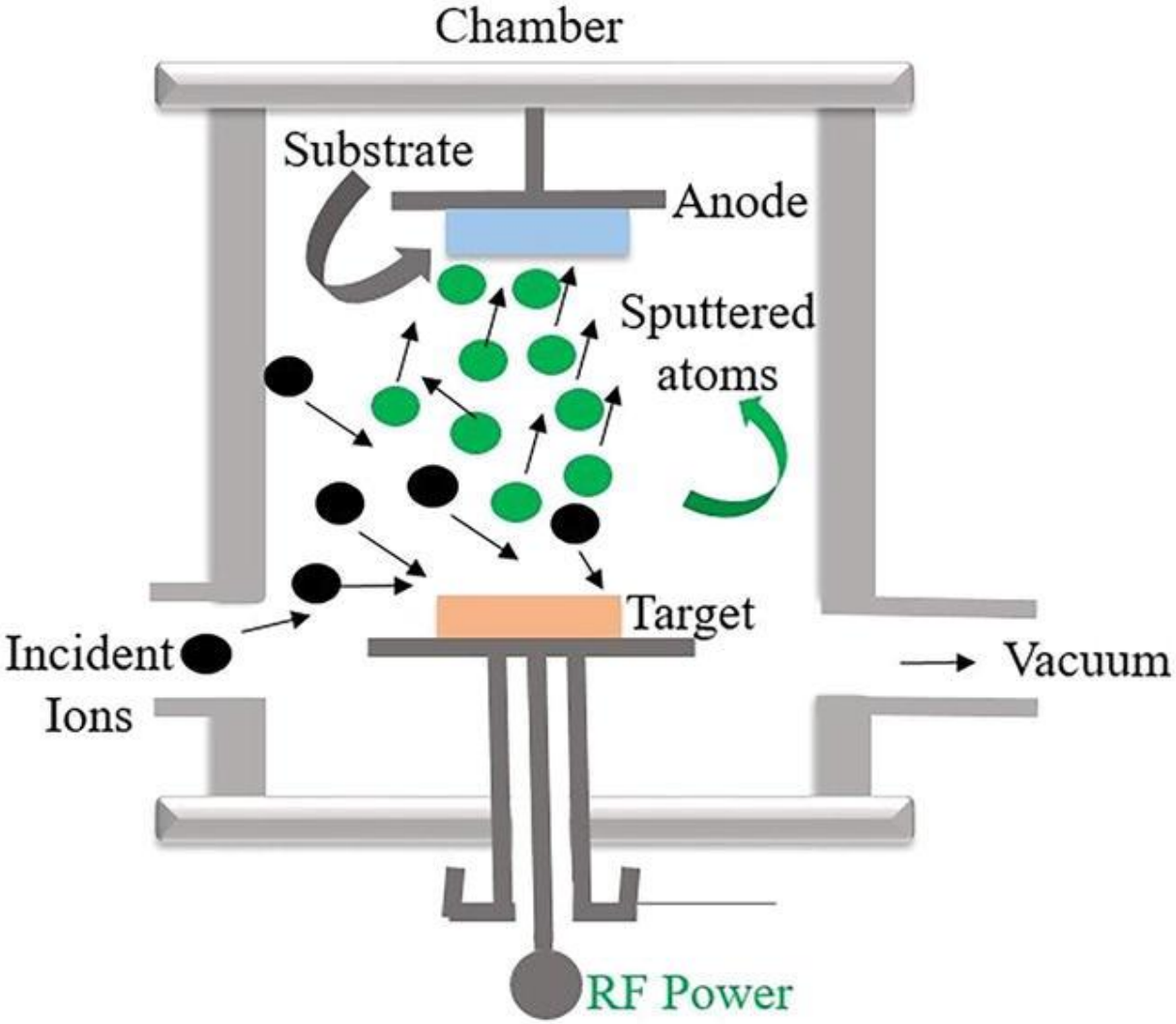


Fig. 3. Different physical and chemical route based deposition methods for preparing CZTS thin films.


Chamber
Substrate
Anode
Sputtered
atoms
Target
Incident
Ions
Vacuum
RF Power

Steps Involved in Sputtering Process

Chamber Preparation

•Substrate is cleaned (ultrasonic cleaning, solvent wash, plasma cleaning).
•Substrate is mounted on the holder inside the sputtering chamber.
•Target material (metal/compound) is fixed on the cathode.

Vacuum Creation

•Chamber is evacuated using:
- Rotary pump (rough vacuum)
- Turbomolecular pump (high vacuum)

•Base pressure typically: **10^{-5} – 10^{-7} Torr**
High vacuum removes contaminants like oxygen and moisture.

Introduction of Sputtering Gas

•Inert gas (usually **Argon**) is introduced.
•Working pressure: **1–10 mTorr**
•Argon is chosen because it is inert and heavy enough for effective momentum transfer.

Plasma Generation

High voltage is applied between:
Target (Cathode, negative)
Chamber walls/Substrate holder (Anode, positive)
Argon gas gets ionized → forms plasma (Ar^+ + electrons).

Sputtering (Target Bombardment)

Positively charged Ar^+ ions accelerate toward the negatively charged target.
Ion bombardment knocks out atoms from the target surface.
This is called momentum transfer process.

Transport of Sputtered Atoms

Ejected atoms travel through vacuum.
They may undergo collisions depending on pressure.
At low pressure → more directional deposition.

Film Deposition on Substrate

Sputtered atoms condense on the substrate.
Thin film nucleation and growth occur.
Parameters influencing film:
Power
Pressure
Substrate temperature
Target-substrate distance
Deposition time

Cooling and Venting

Power is turned off.
Chamber is allowed to cool.
Vacuum is released.
Coated substrate is removed.

Pulsed Laser Ablation Technique

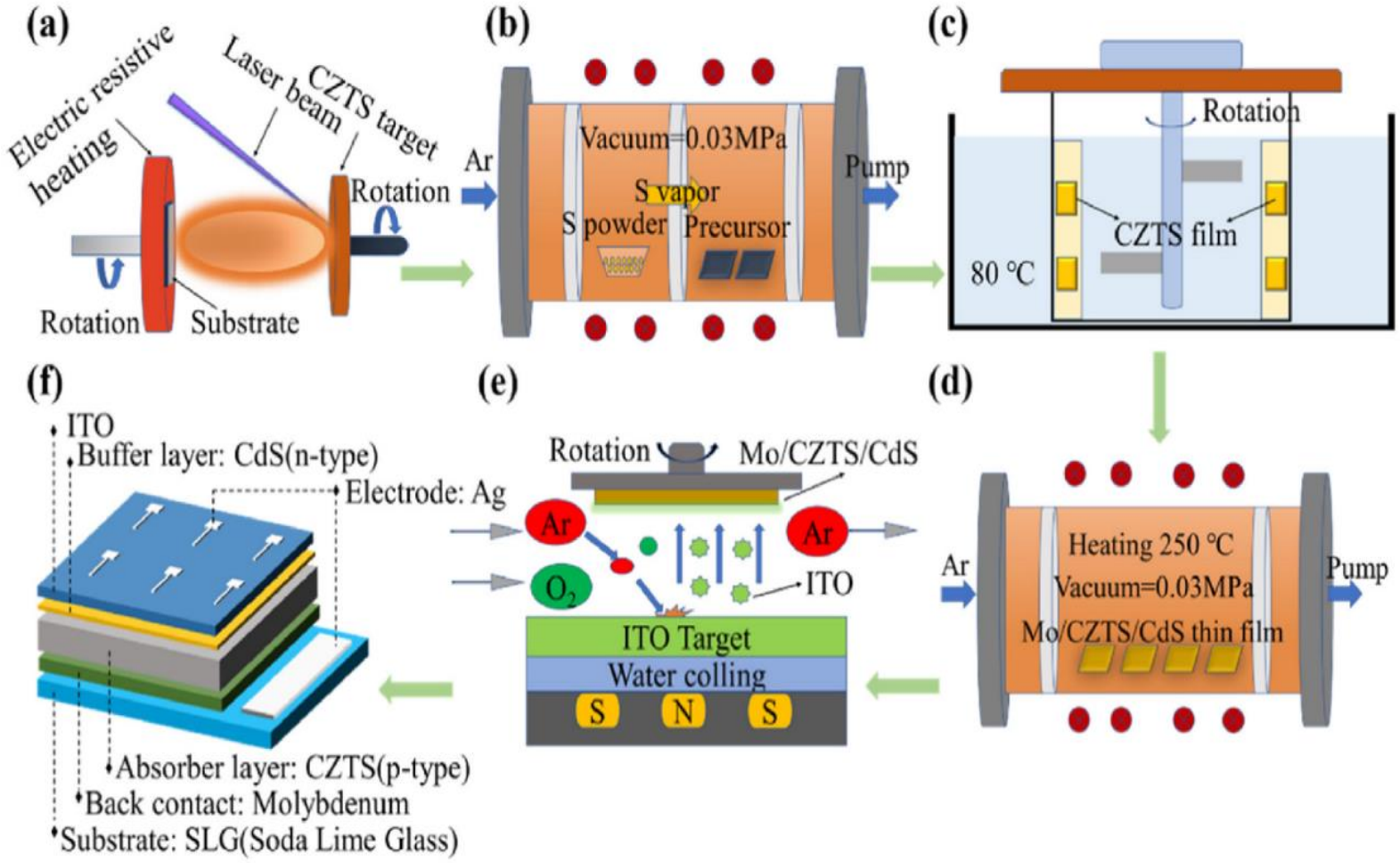


Fig. 5. A graphic representation of the thin film CZTS based solar cell manufacturing process. (a) PLD thin film deposition of the CZTS precursor; (b) post-sulfurization annealing to produce the CZTS film; (c) CBD thin film deposition of the CdS buffer layer; (d) post-heat treatment of CZTS/CdS heterojunction; (e) magnetron sputtering thin film deposition of the ITO layer; and (f) final structural illustration of the thin film CZTS based solar cell. Adapted with permission from Ref. 151. Copyright (2021), Elsevier.

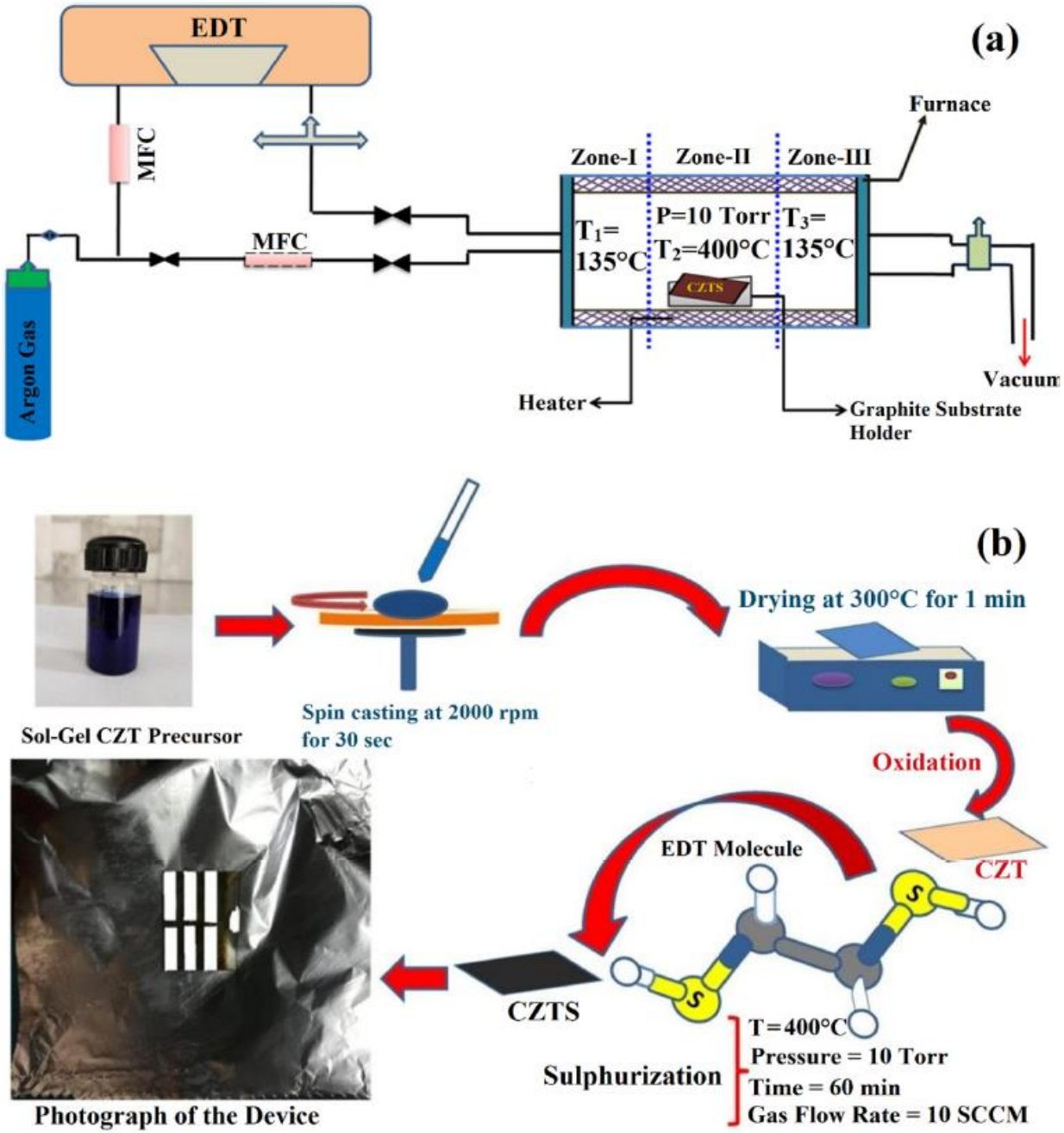


Fig. 8. (a) Schematic illustration of horizontal tubular reactor cast-off for sulfurization, (b) schematic process to develop CZTS thin films and the devices. Adapted with permission from Ref. 214. Copyright (2022), Elsevier.

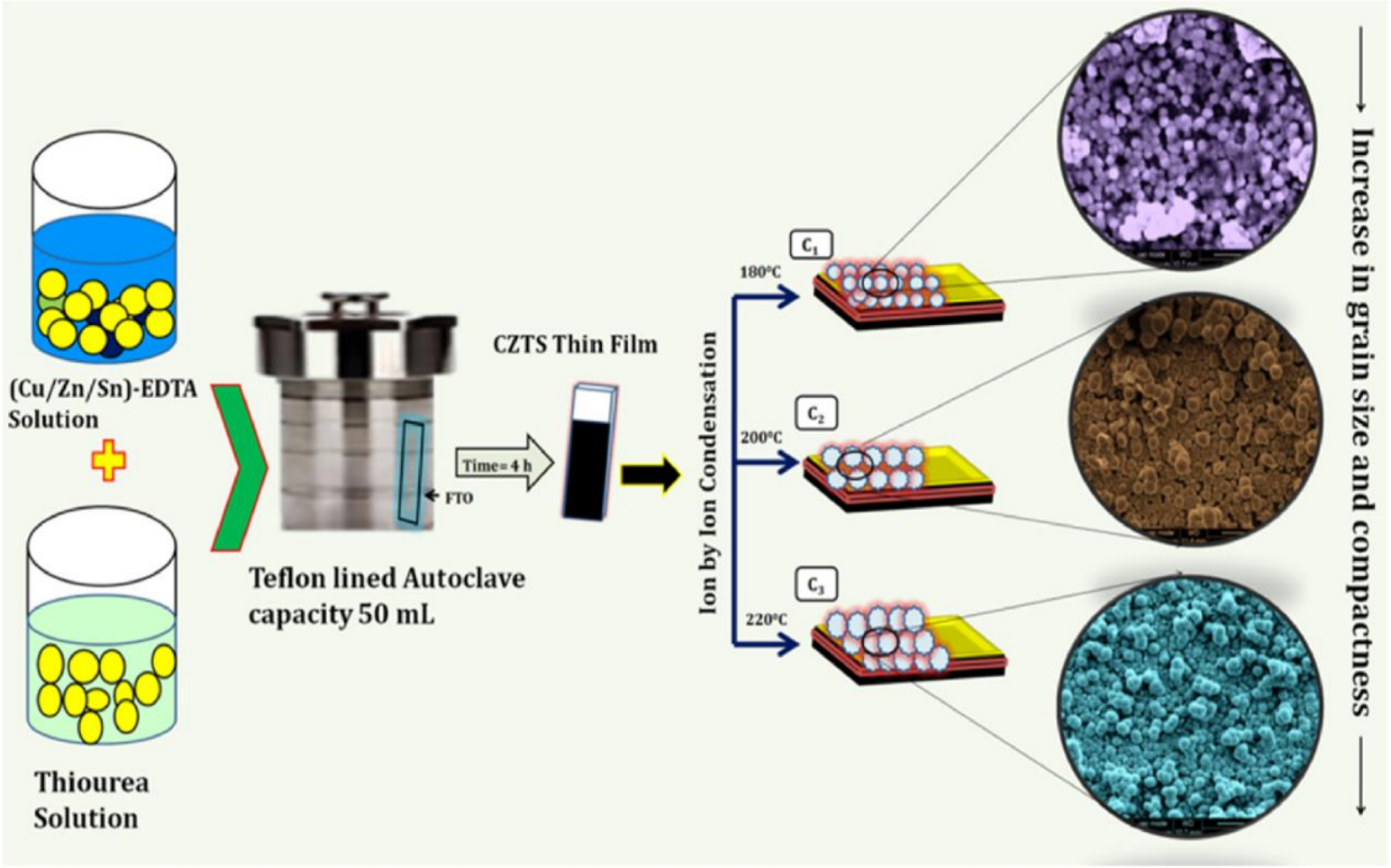
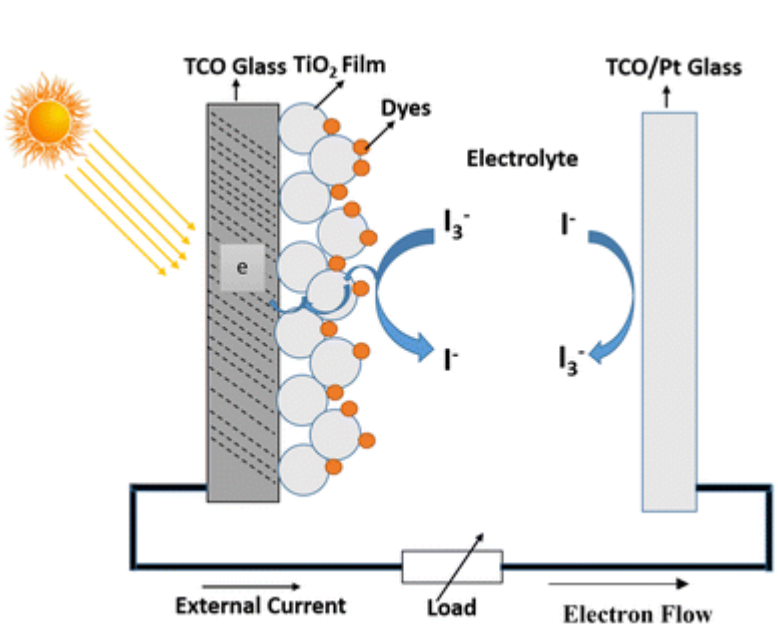


Fig. 14. Feasible development mechanism for synthesis of the thin CZTS film via modified one-step hydrothermal route. Adapted with permission from Ref. 304. Copyright (2021), American Chemical Society.

Dye-sensitized Solar Cells



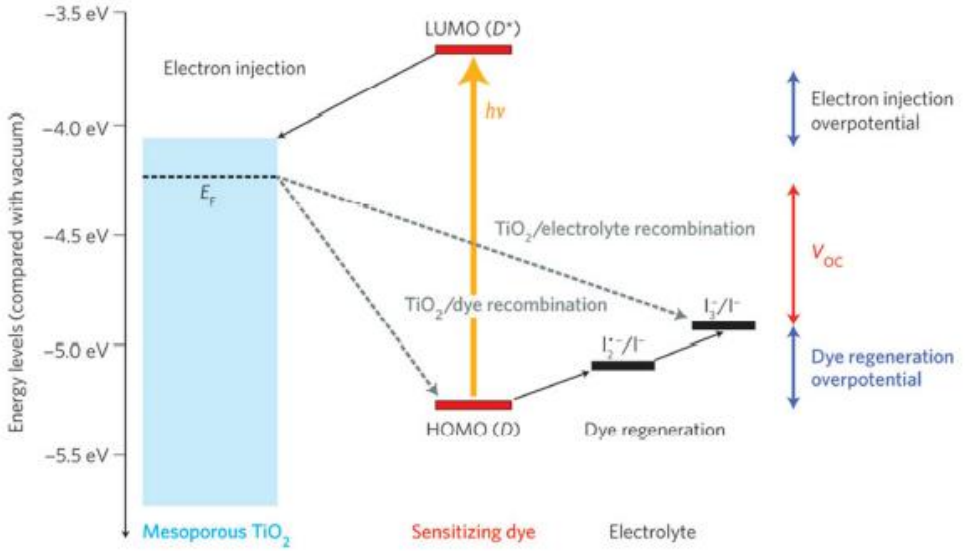
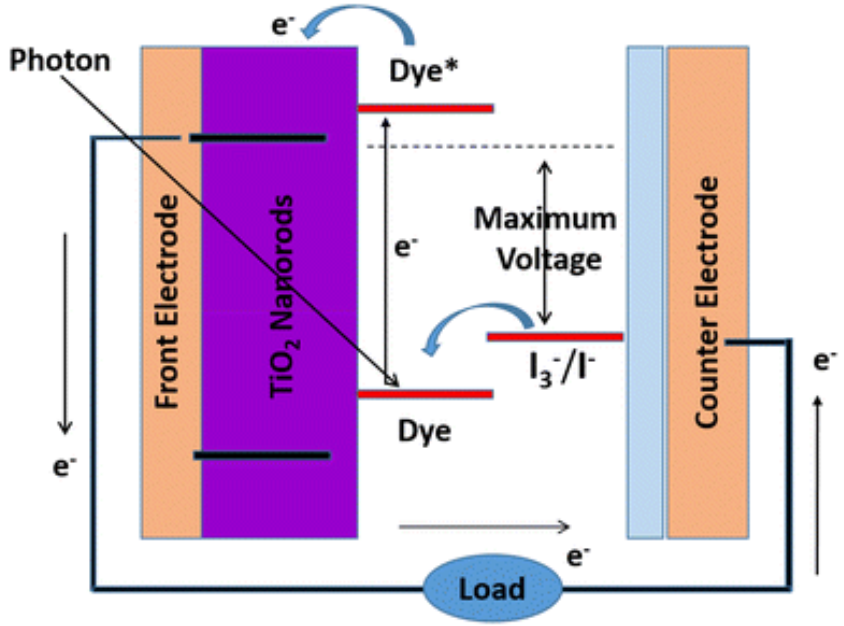
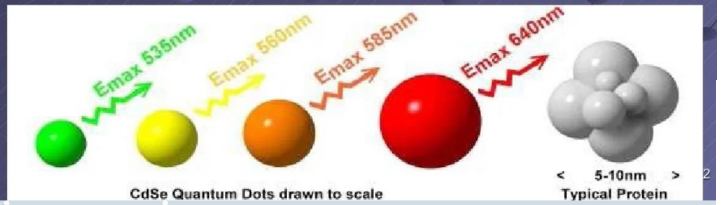


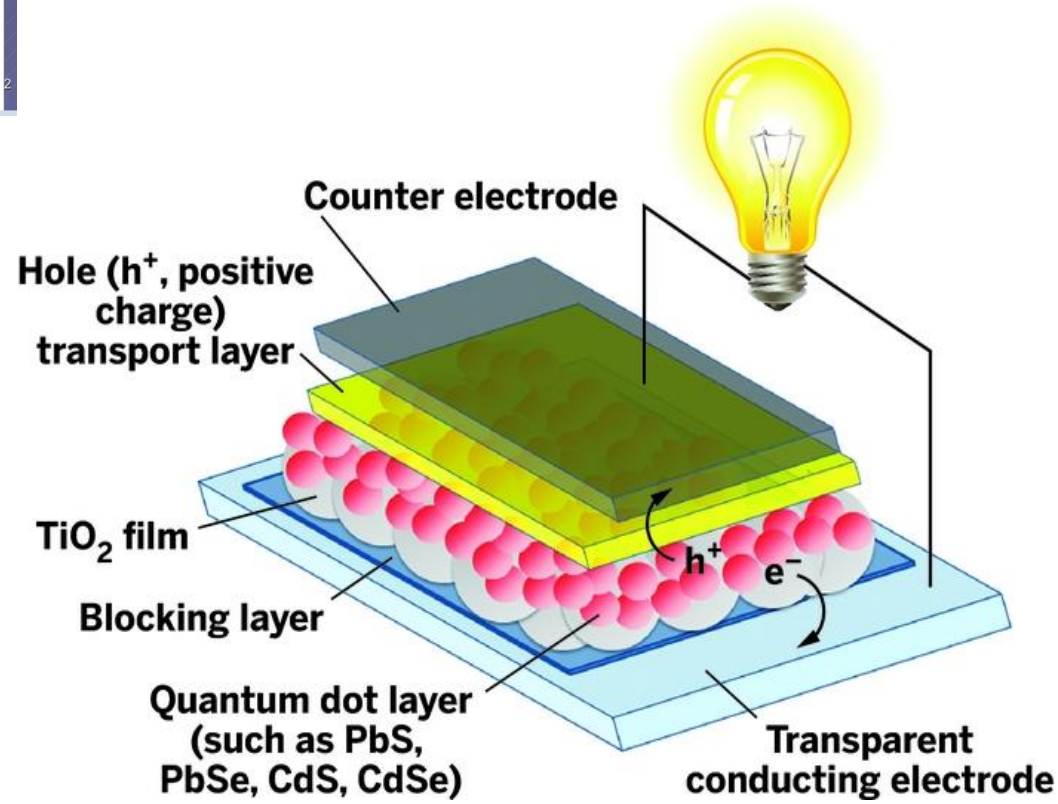
Fig. 8. Dye-sensitized solar cell device schematic and operation.



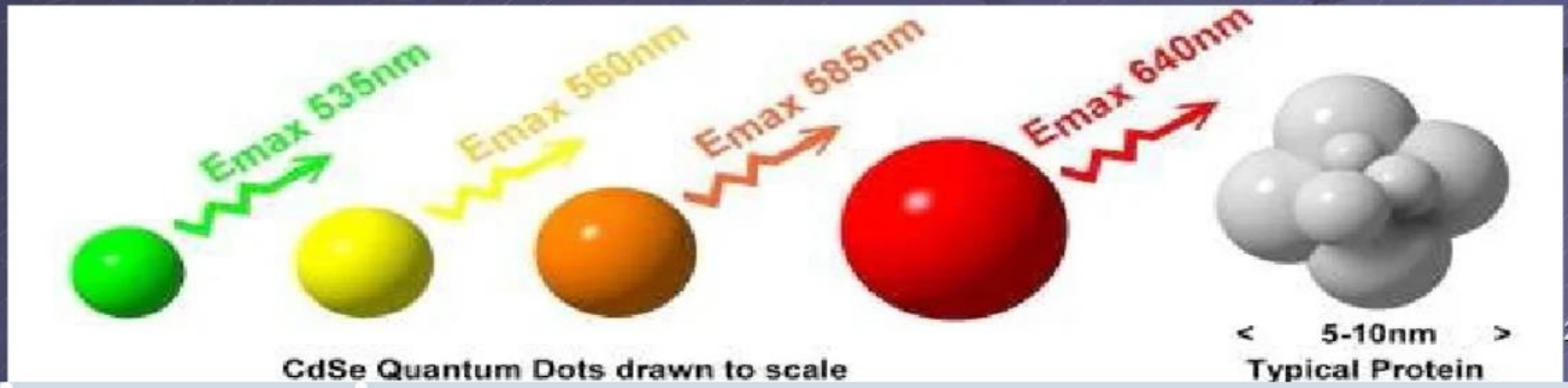
Quantum Dot solar Cells

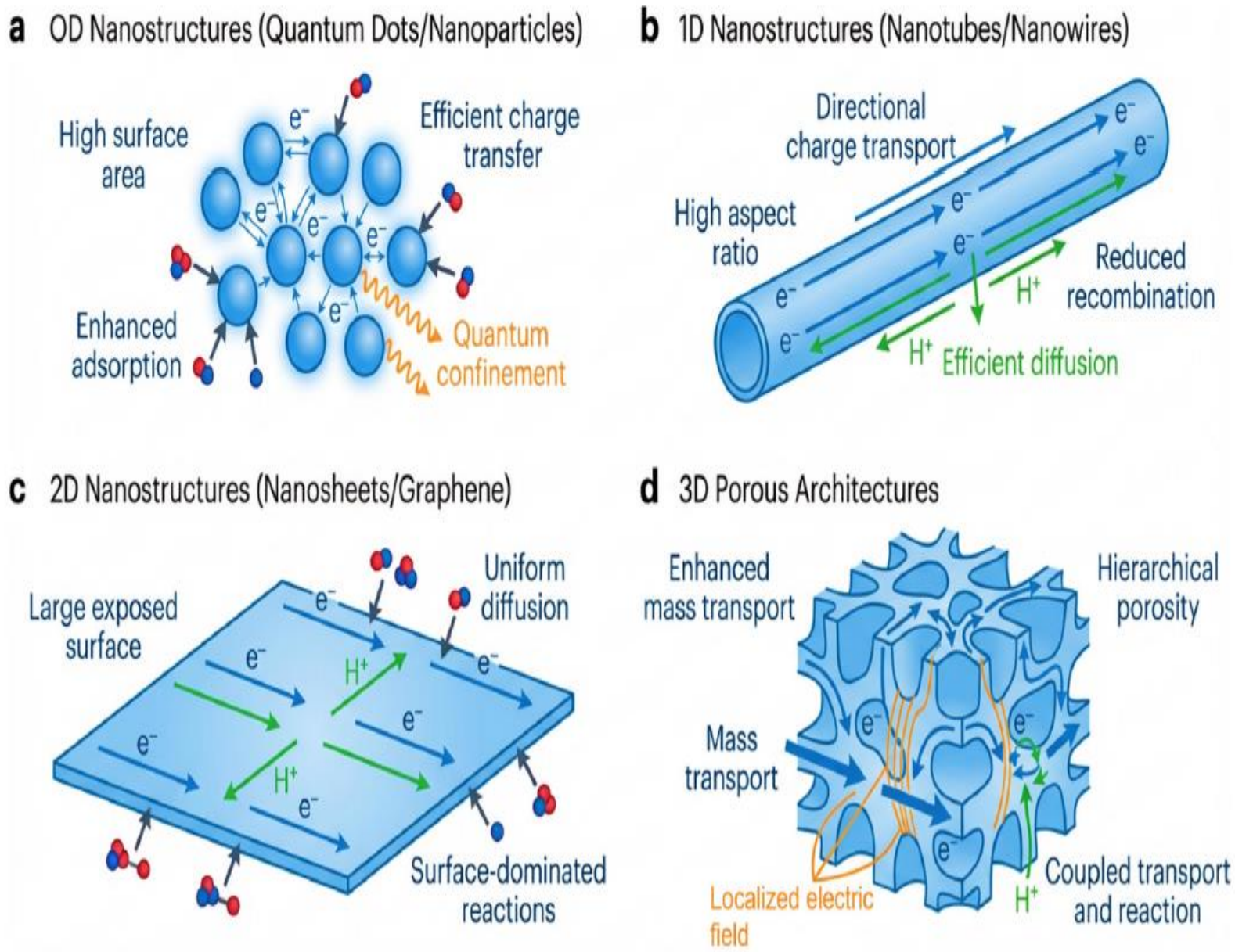
- Quantum dots are semiconductors whose excitons are confined in all three dimensions of space.
- Quantum dots have properties combined between
  - Those of bulk semiconductors
  - Those of atoms
- Different methods to create quantum dots.
- Multiple applications.

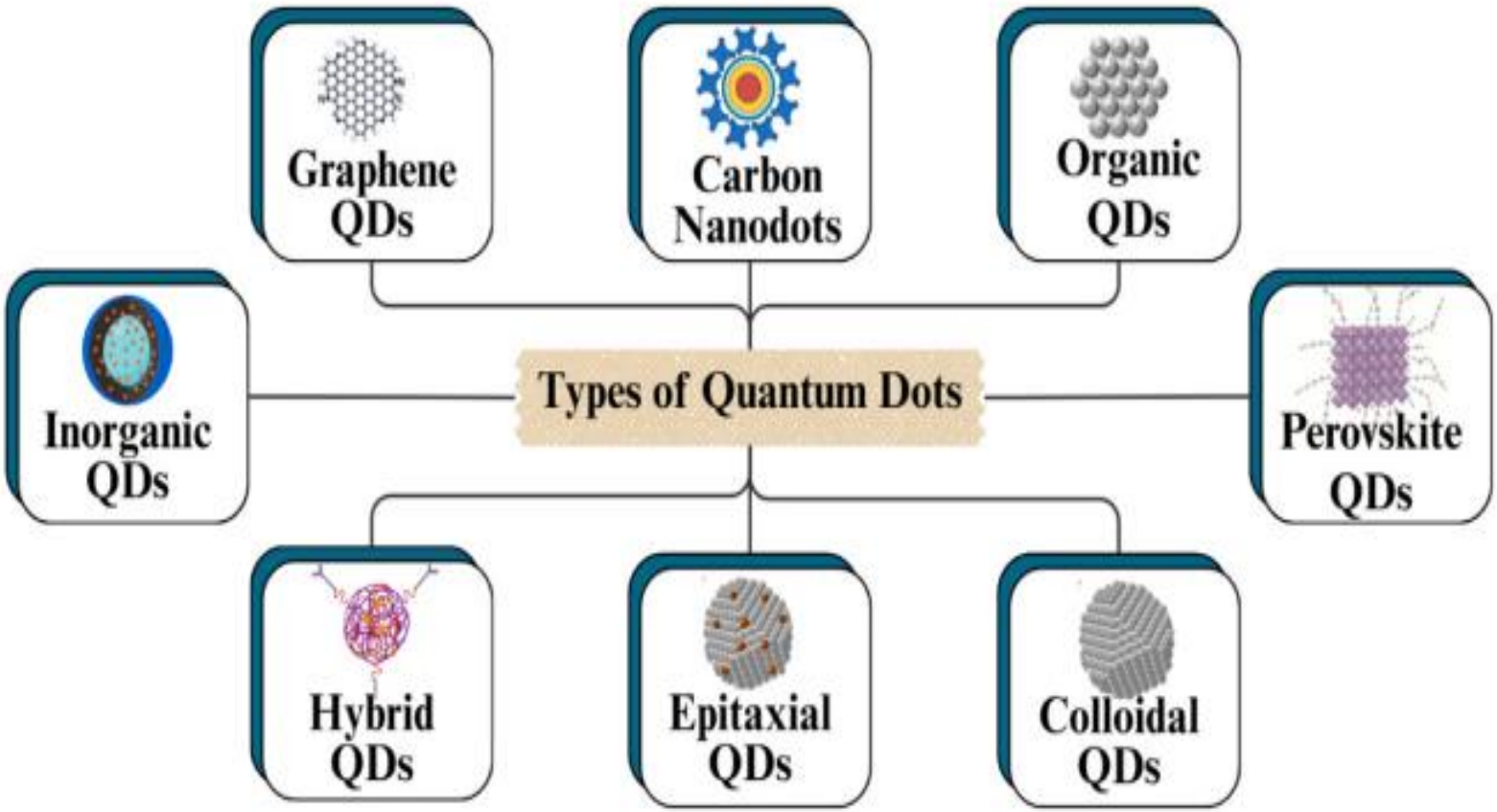


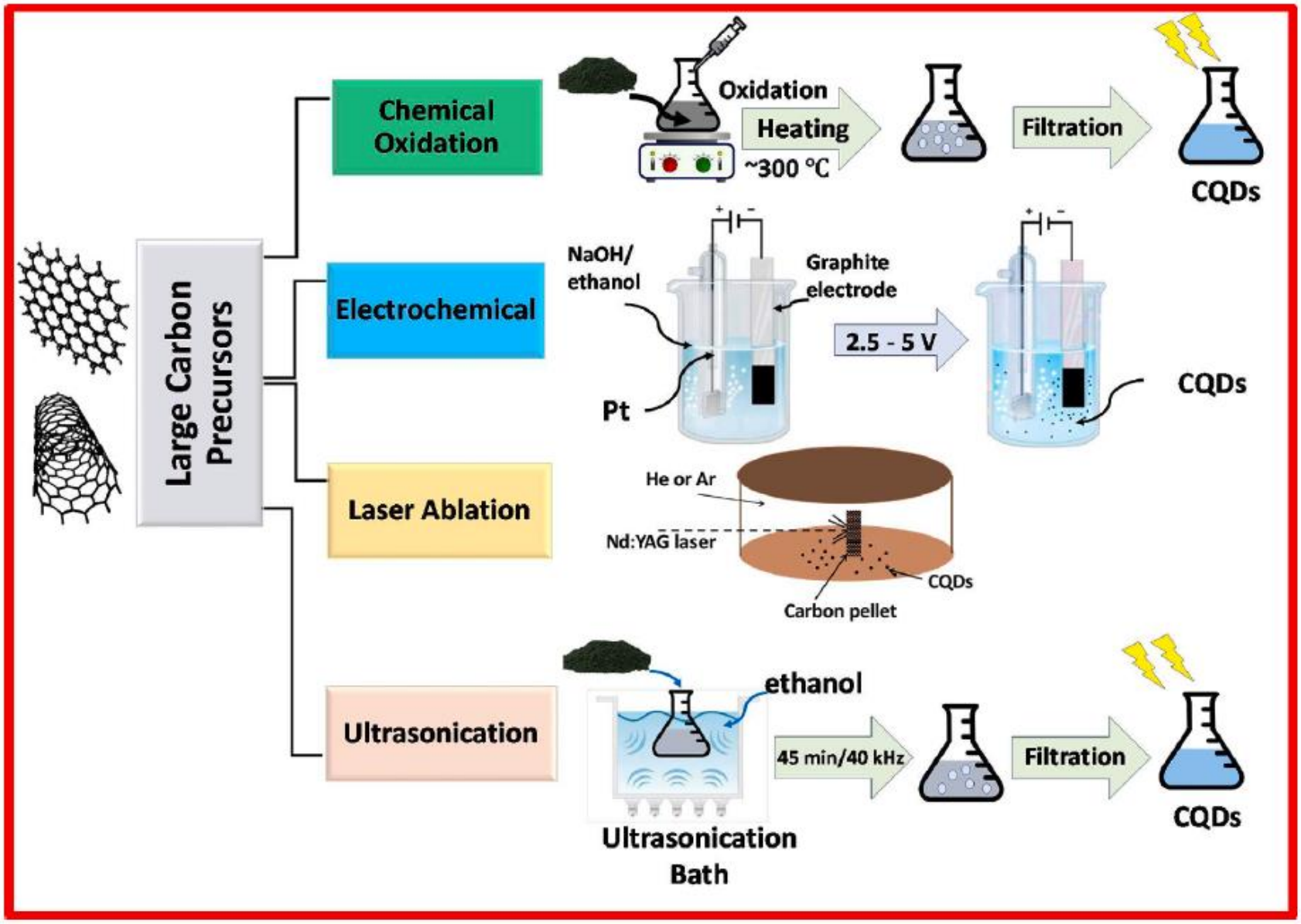


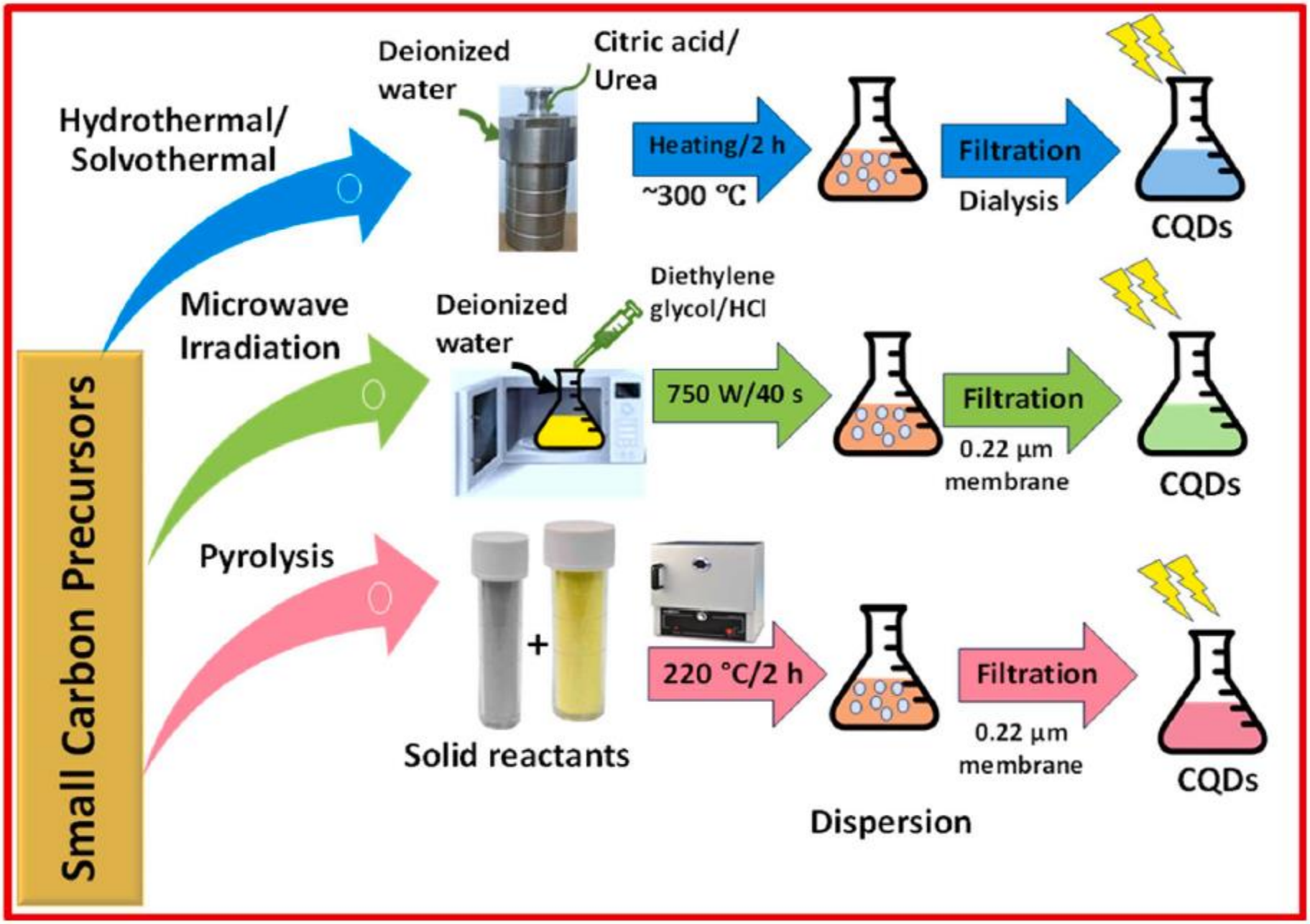
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- Different methods to create quantum dots.
- Multiple applications.

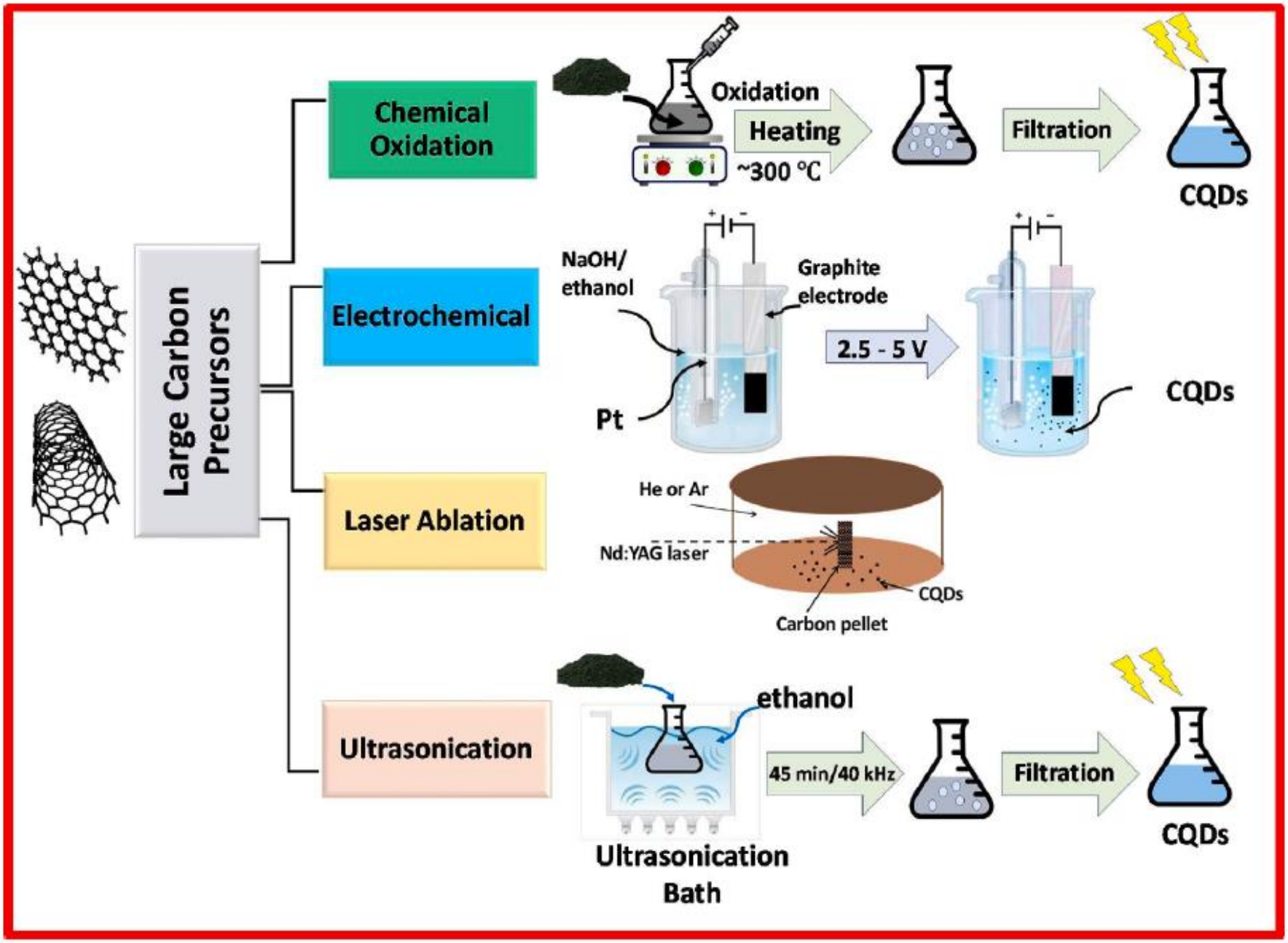


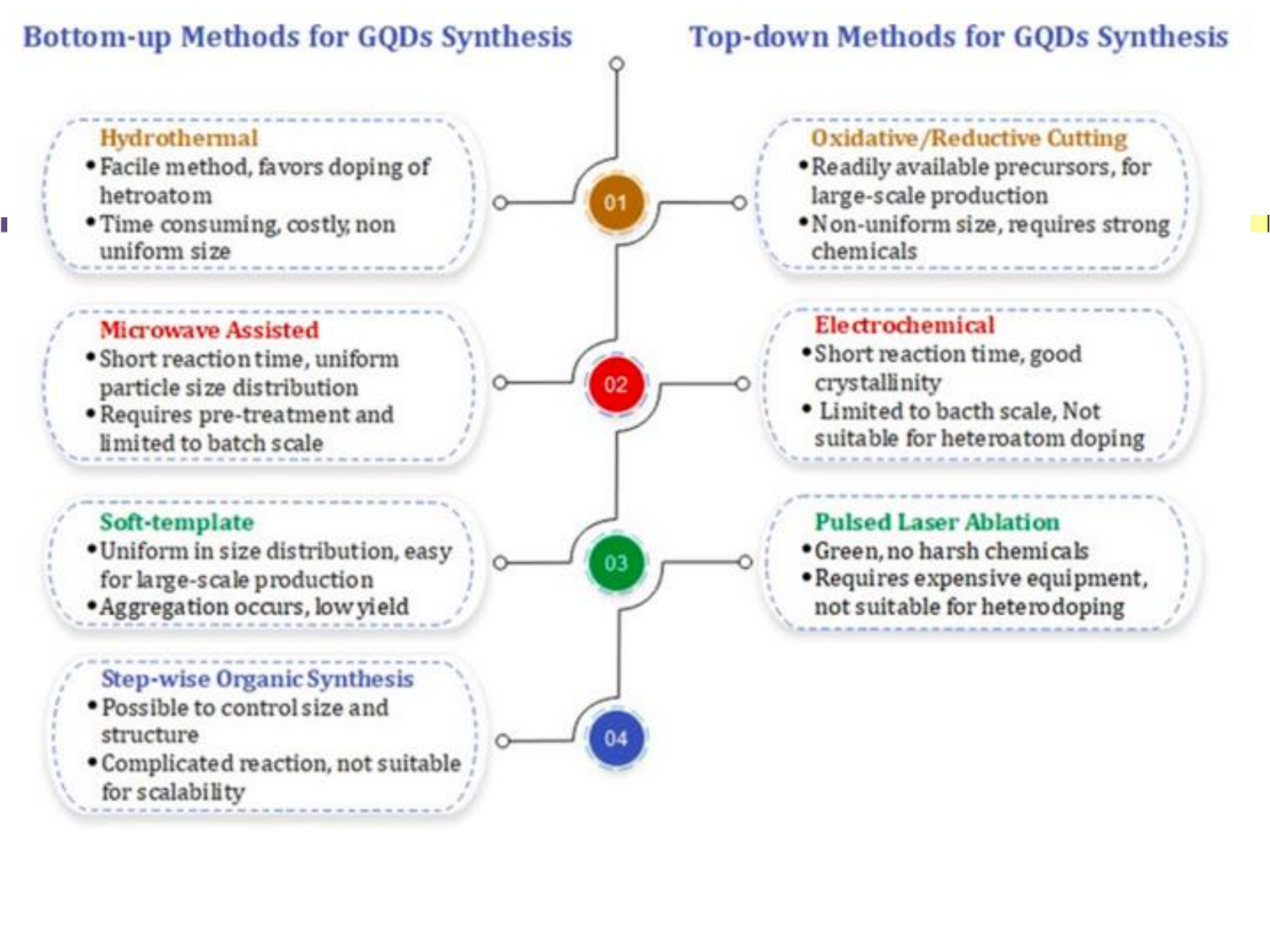

a OD Nanostructures (Quantum Dots/Nanoparticles)
High surface area
Efficient charge transfer
Enhanced adsorption
Quantum confinement
b 1D Nanostructures (Nanotubes/Nanowires)
Directional charge transport
High aspect ratio
Reduced recombination
Efficient diffusion
c 2D Nanostructures (Nanosheets/Graphene)
Large exposed surface
Uniform diffusion
Surface-dominated reactions
d 3D Porous Architectures
Enhanced mass transport
Hierarchical porosity
Mass transport
Localized electric field
Coupled transport and reaction

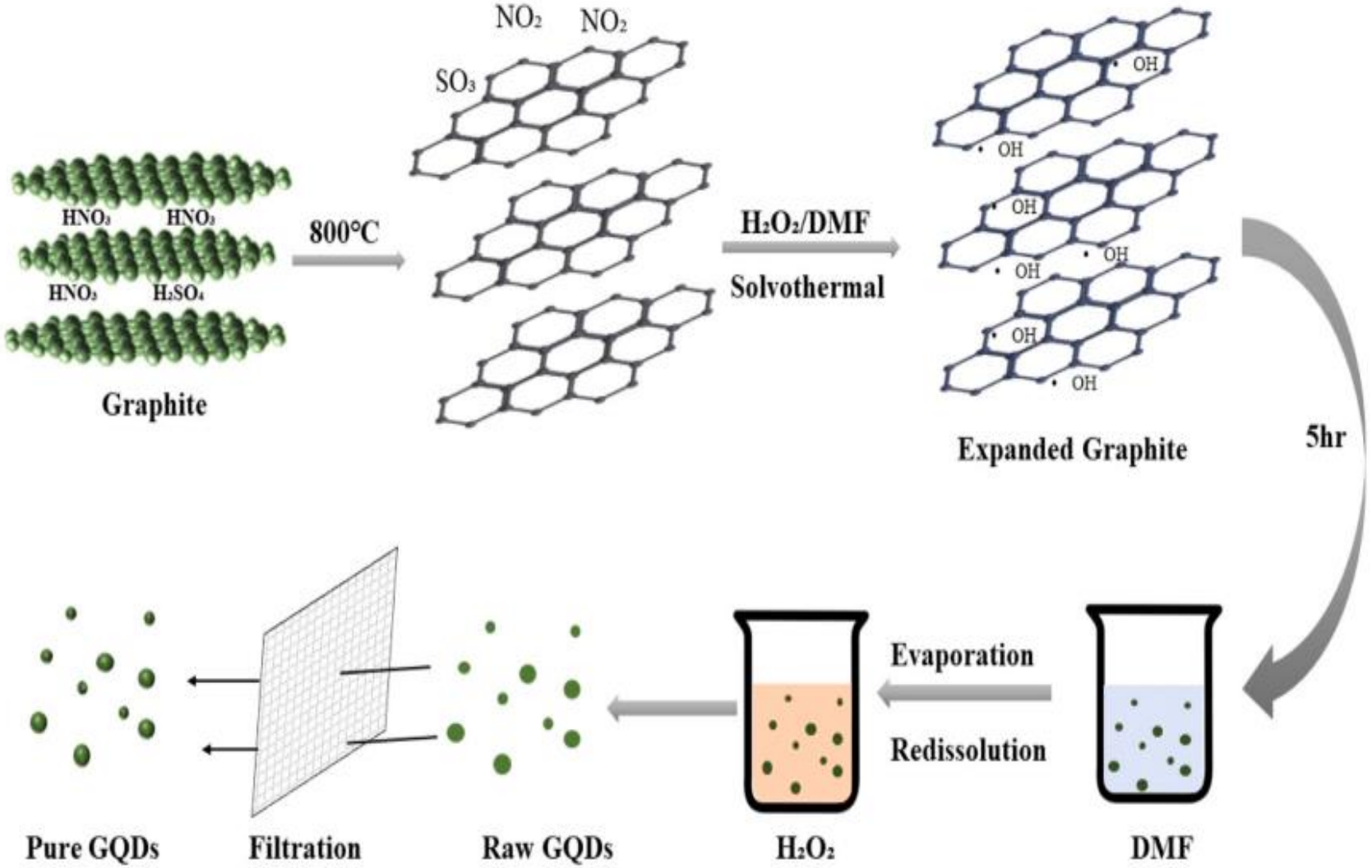

Graphene QDs
Carbon Nanodots
Organic QDs
Inorganic QDs
Types of Quantum Dots
Perovskite QDs
Hybrid QDs
Epitaxial QDs
Colloidal QDs

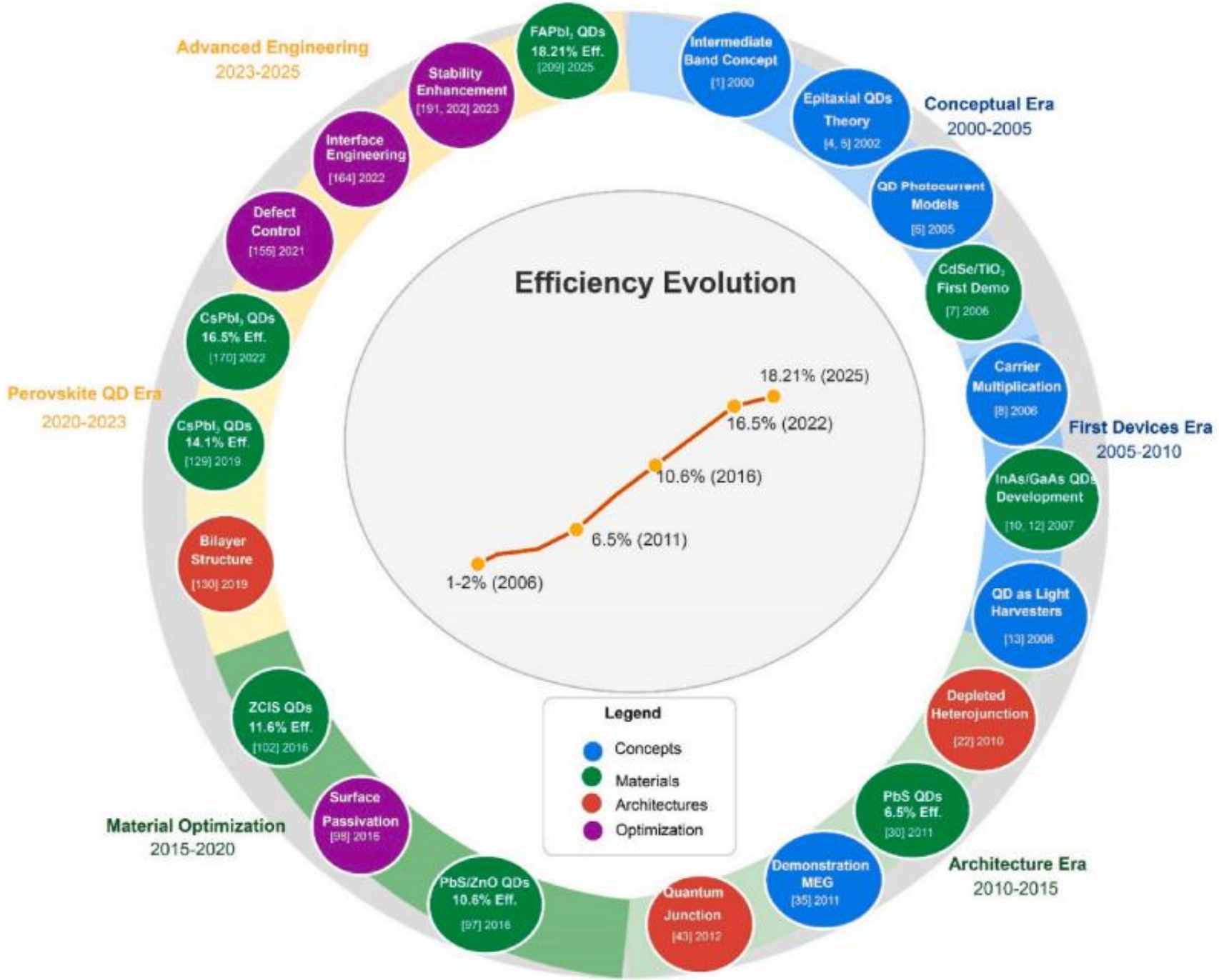

Large Carbon Precursors
Chemical Oxidation
Oxidation
Heating
~300 °C
Filtration
CQDs
Electrochemical
NaOH/ ethanol
Graphite electrode
Pt
2.5 - 5 V
CQDs
Laser Ablation
He or Ar
Nd:YAG laser
CQDs
Carbon pellet
Ultrasonication
ethanol
45 min/40 kHz
Filtration
CQDs
Ultrasonication Bath


Small Carbon Precursors
Hydrothermal/
Solvothermal
Deionized
water
Citric acid/
Urea
Heating/2 h
~300 °C
Filtration
Dialysis
CQDs
Microwave
Irradiation
Deionized
water
Diethylene
glycol/HCl
750 W/40 s
Filtration
0.22 μm
membrane
CQDs
Pyrolysis
+
Solid reactants
220 °C/2 h
Filtration
0.22 μm
membrane
CQDs
Dispersion

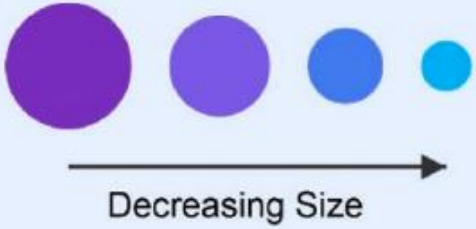

Large Carbon Precursors
Chemical Oxidation
Oxidation
Heating
~300 °C
Filtration
CQDs
Electrochemical
NaOH/ ethanol
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CQDs
Laser Ablation
He or Ar
Nd:YAG laser
CQDs
Carbon pellet
Ultrasonication
ethanol
45 min/40 kHz
Filtration
CQDs
Ultrasonication Bath


Bottom-up Methods for GQDs Synthesis
Top-down Methods for GQDs Synthesis
Hydrothermal
• Facile method, favors doping of hetroatom
• Time consuming, costly, non uniform size
01
Oxidative/Reductive Cutting
• Readily available precursors, for large-scale production
• Non-uniform size, requires strong chemicals
Microwave Assisted
• Short reaction time, uniform particle size distribution
• Requires pre-treatment and limited to batch scale
02
Electrochemical
• Short reaction time, good crystallinity
• Limited to bacth scale, Not suitable for heteroatom doping
Soft-template
• Uniform in size distribution, easy for large-scale production
• Aggregation occurs, low yield
03
Pulsed Laser Ablation
• Green, no harsh chemicals
• Requires expensive equipment, not suitable for heterodoping
Step-wise Organic Synthesis
• Possible to control size and structure
• Complicated reaction, not suitable for scalability
04

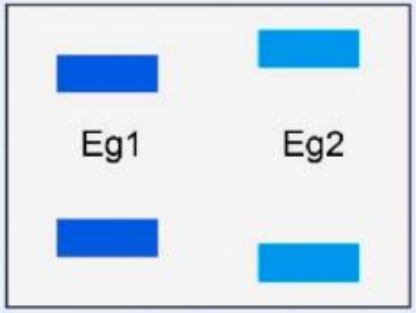

HNO3
HNO3
HNO3
H2SO4
Graphite
800°C
NO2
NO2
SO3
H2O2/DMF
Solvothermal
OH
OH
OH
OH
OH
OH
Expanded Graphite
5hr
DMF
Evaporation
Redissolution
H2O2
Raw GQDs
Filtration
Pure GQDs


Advanced Engineering
2023-2025
FAPbI3 QDs
18.21% Eff.
[209] 2025
Stability Enhancement
[191, 202] 2023
Interface Engineering
[164] 2022
Defect Control
[155] 2021
Perovskite QD Era
2020-2023
CsPbI3 QDs
16.5% Eff.
[170] 2022
CsPbI3 QDs
14.1% Eff.
[129] 2019
Bilayer Structure
[130] 2019
Material Optimization
2015-2020
ZCIS QDs
11.6% Eff.
[102] 2016
Surface Passivation
[98] 2015
PbS/ZnO QDs
10.6% Eff.
[97] 2016
Quantum Junction
[43] 2012
Architecture Era
2010-2015
Demonstration MEG
[35] 2011
PbS QDs
6.5% Eff.
[30] 2011
Depleted Heterojunction
[22] 2010
QD as Light Harvesters
[13] 2008
First Devices Era
2005-2010
InAs/GaAs QDs
Development
[10, 12] 2007
Carrier Multiplication
[8] 2006
CdSe/TiO2
First Demo
[7] 2006
QD Photocurrent Models
[6] 2005
Conceptual Era
2000-2005
Epitaxial QDs Theory
[4, 5] 2002
Intermediate Band Concept
[1] 2000
Efficiency Evolution
18.21% (2025)
16.5% (2022)
10.6% (2016)
6.5% (2011)
1-2% (2006)
Legend
Concepts
Materials
Architectures
Optimization

Bandgap Tunability



Decreasing Size





Matches Solar Spectrum

- Spectral matching
- Multi-junction devices
- Intermediate band cells

Multiple Exciton Generation



hv > 2Eg





Charge Extraction

Conventional: 1 photon → 1 e-h pair

QD with MEG: 1 photon → multiple e-h pairs

- Enhanced photocurrent
- Better high-energy photon use
- Beyond Shockley-Queisser limit

Solution Processability

Spin Coating

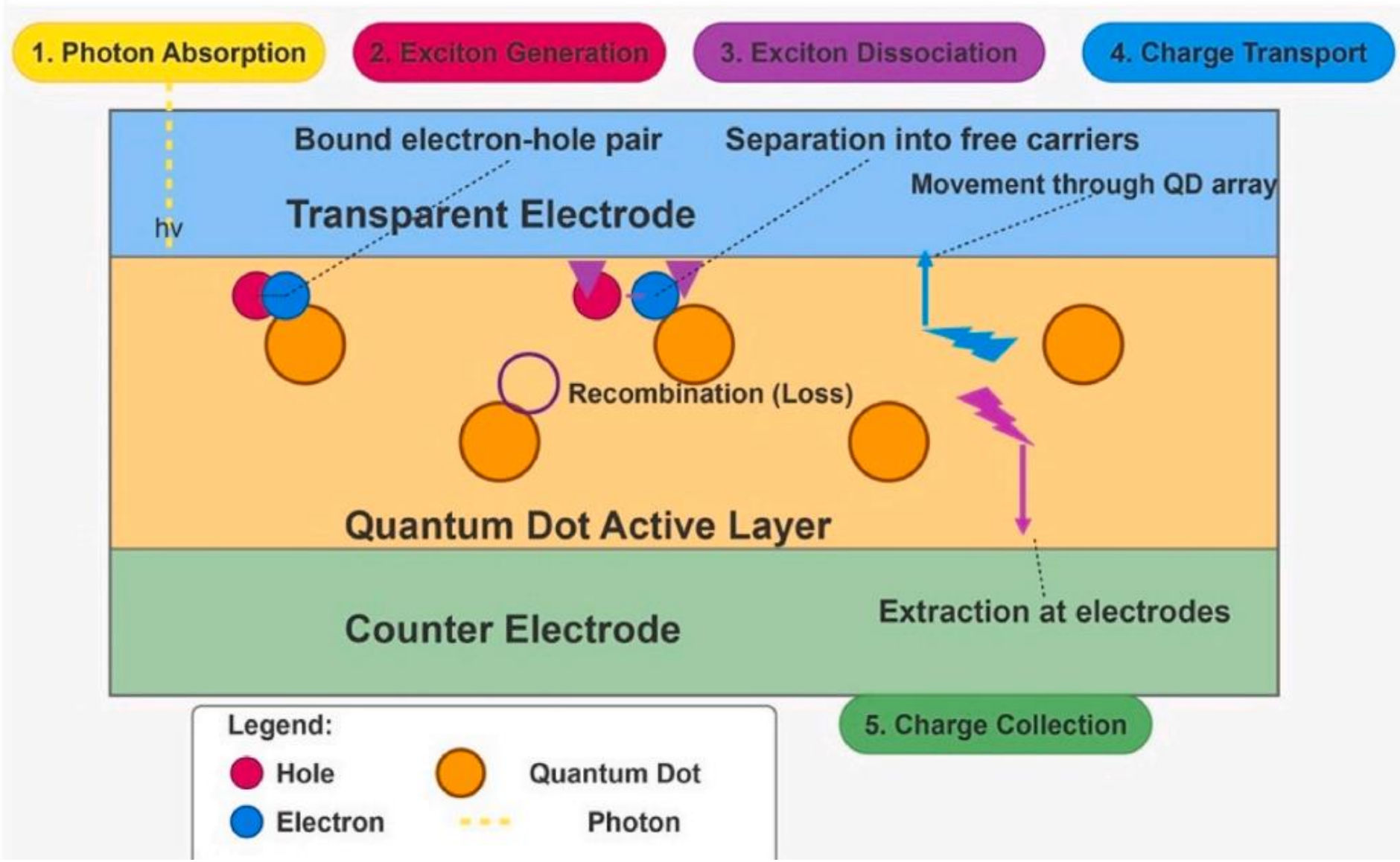
Blade Coating

Spray Deposition

Inkjet Printing

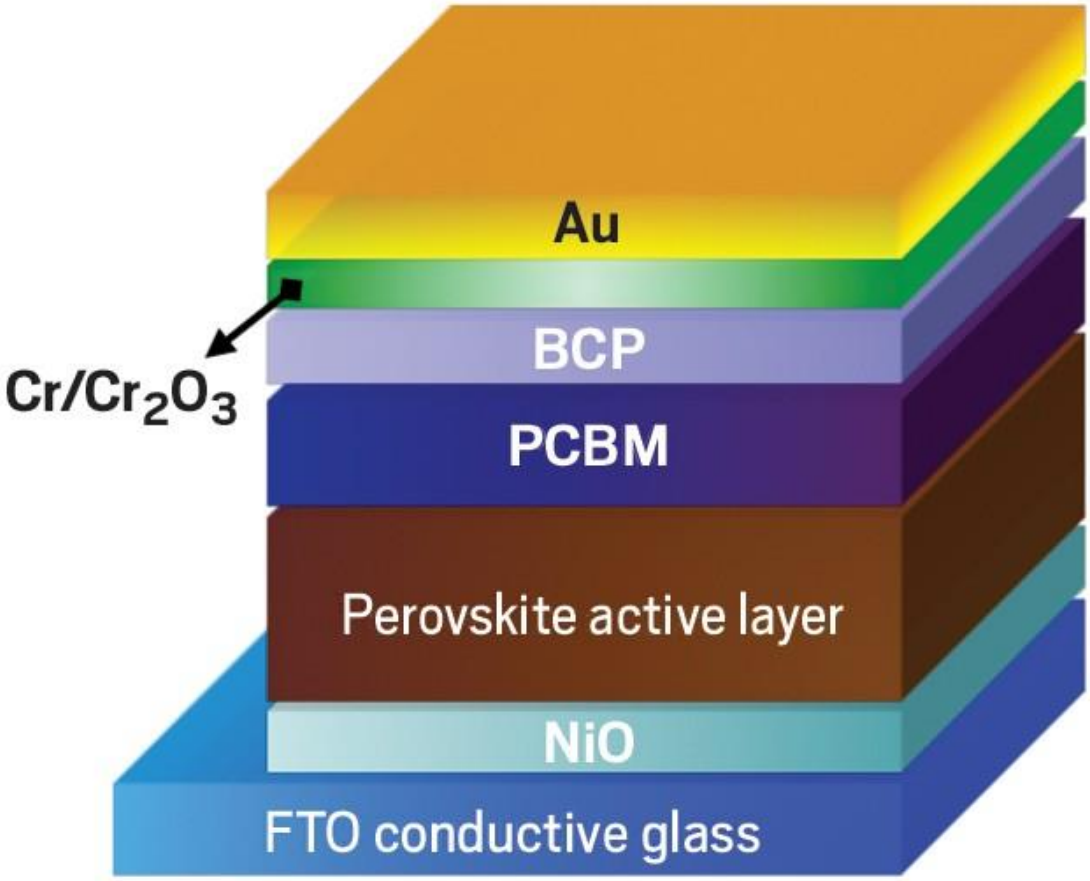
Benefits:

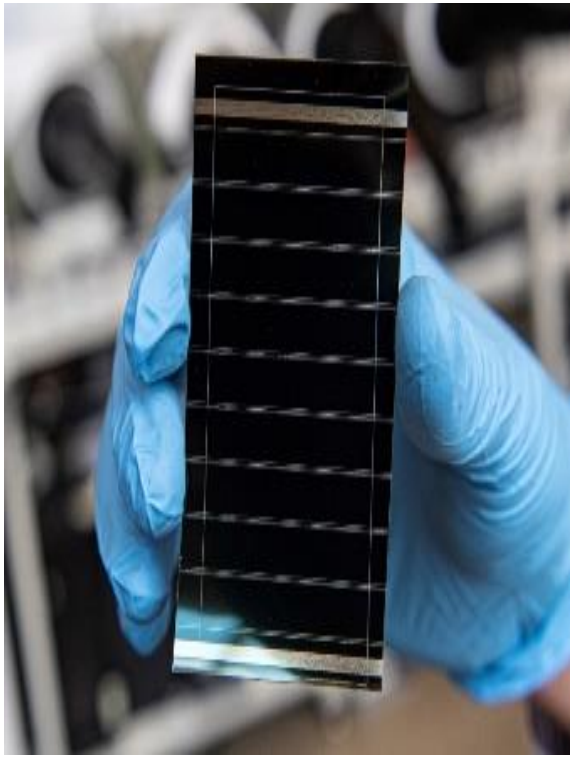
- Low-cost manufacturing
- Ambient processing
- Flexible substrates
- Large-area production


1. Photon Absorption
2. Exciton Generation
3. Exciton Dissociation
4. Charge Transport
Bound electron-hole pair
Separation into free carriers
Movement through QD array
hν
Transparent Electrode
Recombination (Loss)
Quantum Dot Active Layer
Counter Electrode
Extraction at electrodes
5. Charge Collection
Legend:
Hole
Quantum Dot
Electron
Photon

Perovskite Solar Cells

$CaTiO_3$ ($^{XII}A^{2+\,VI}B^{4+}X^{2-}{}_3$), known as the perovskite structure





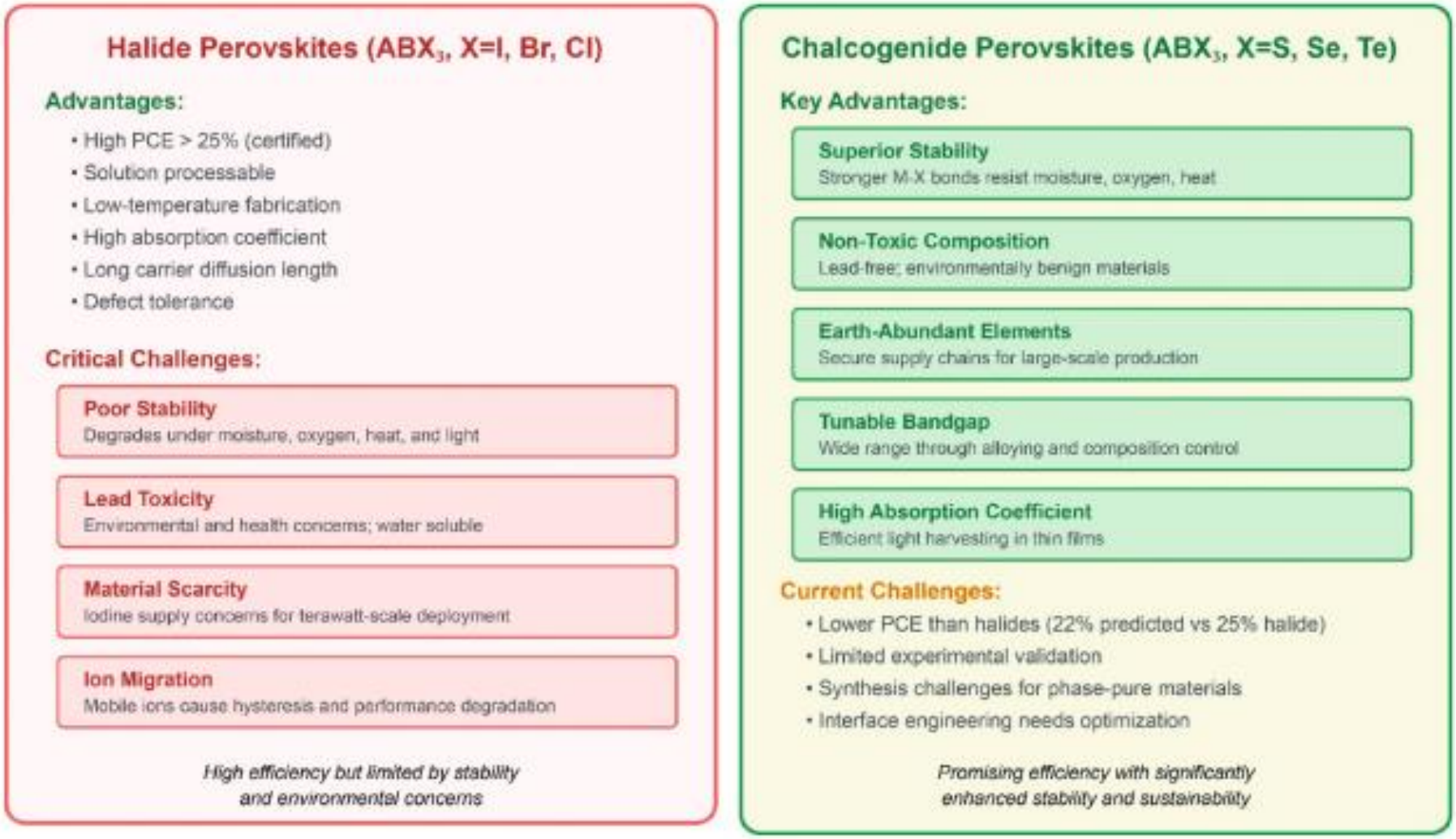


Fig. 2. Comparative analysis of halide versus chalcogenide perovskites.

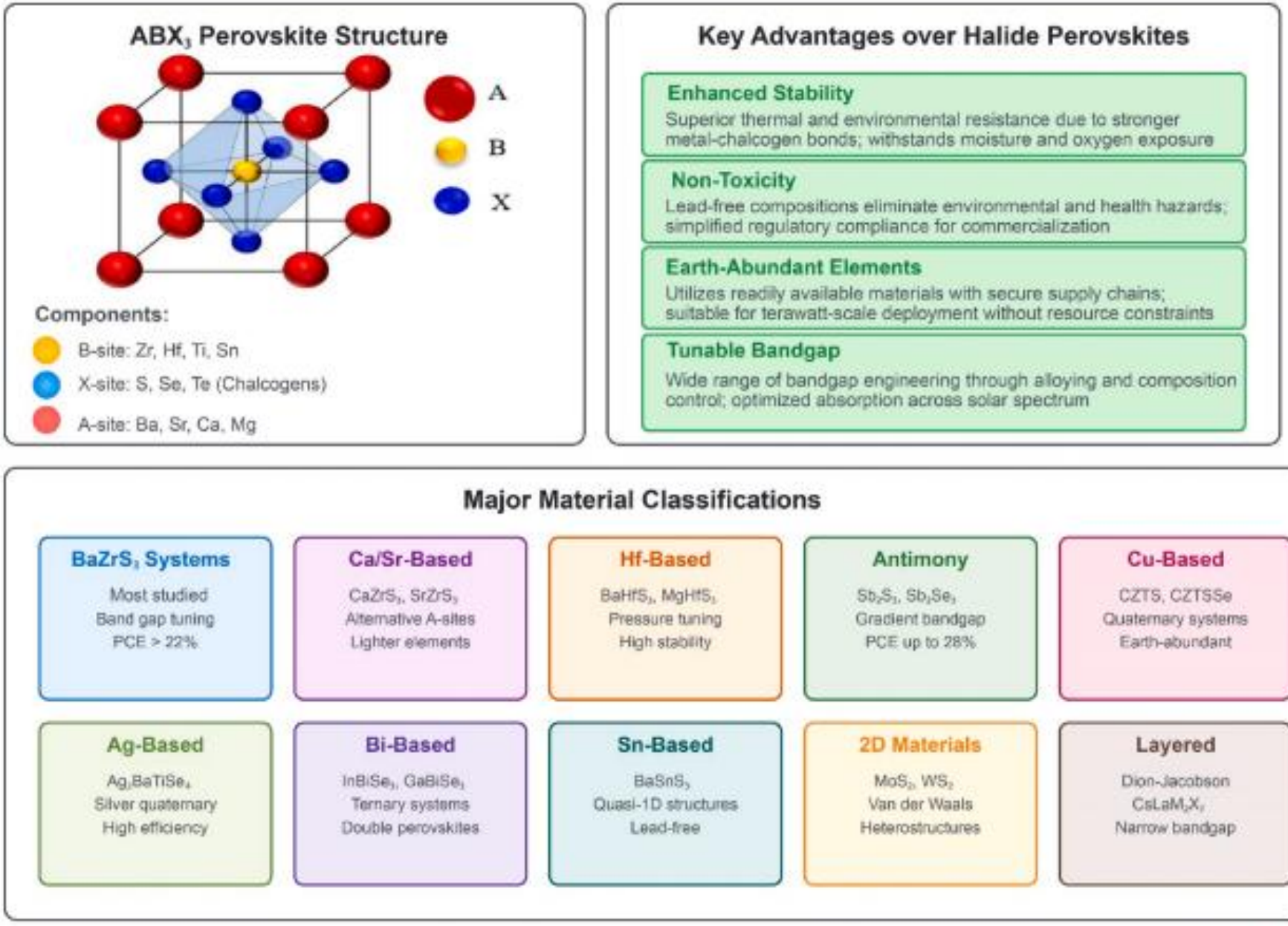


Fig. 3. Crystal Structure and Material Classification of Chalcogenide Perovskites. (ABX_3 Perovskite structure is reprinted from Tanusri Saha-Dasgupta (2020), *Mater. Res. Express* 7 014003 [8] licensed under CC BY 4.0.).

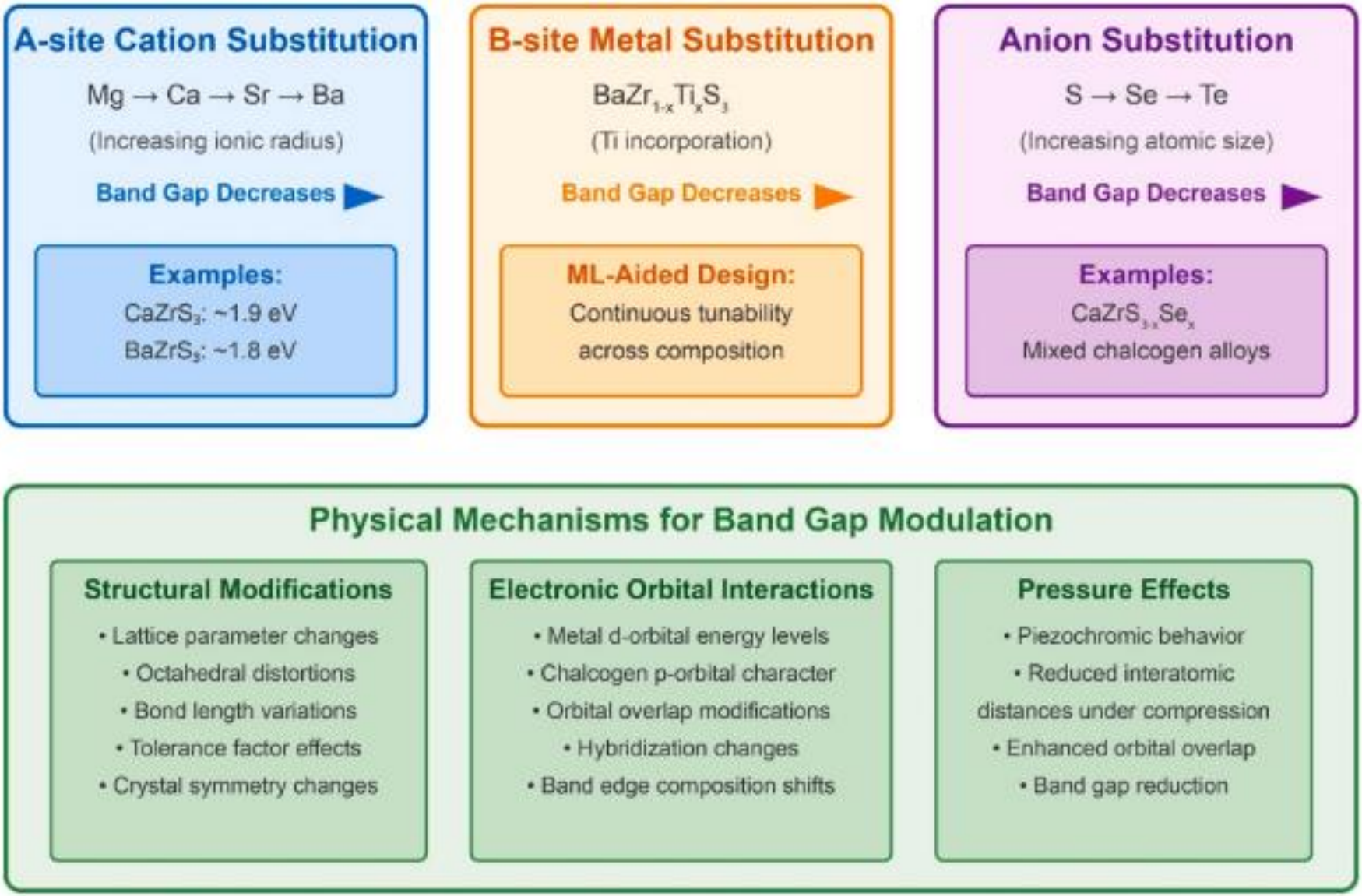


Fig. 5. Band gap tunability mechanisms in chalcogenide perovskites.

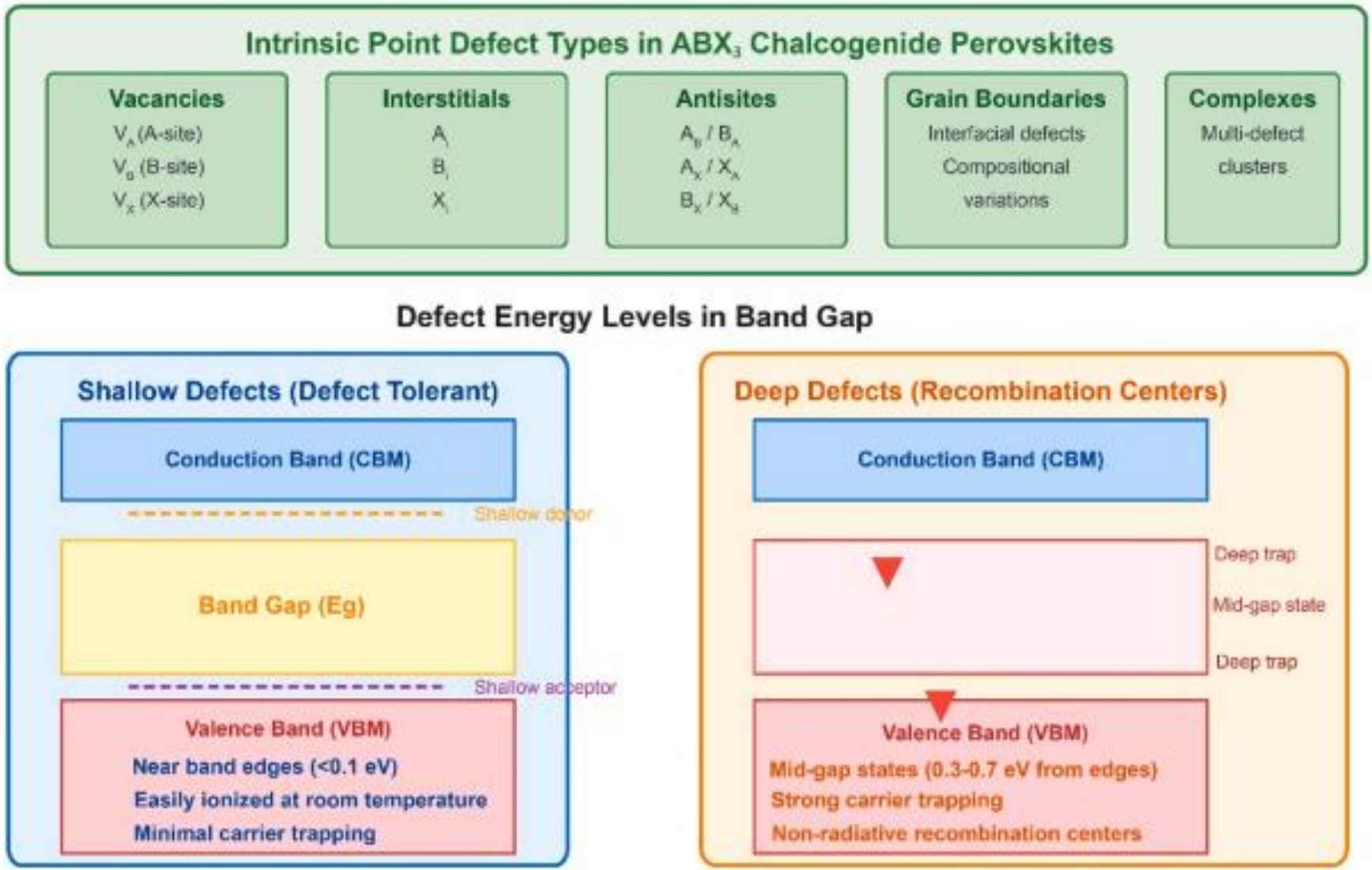
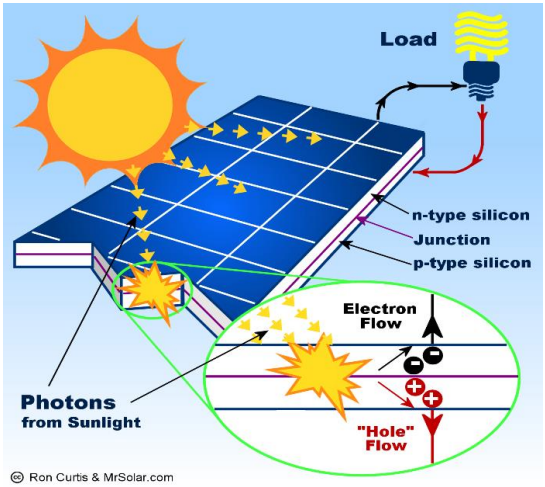


Fig. 6. Defect tolerance in chalcogenide perovskites.

Possible ways of Solar Energy Conversion



- Solar cells
- 22 % max efficiency

Solar

to

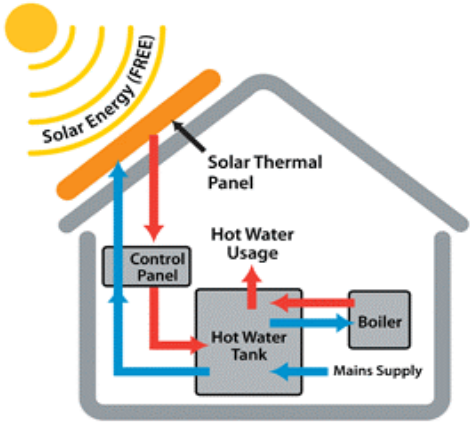
Electricity

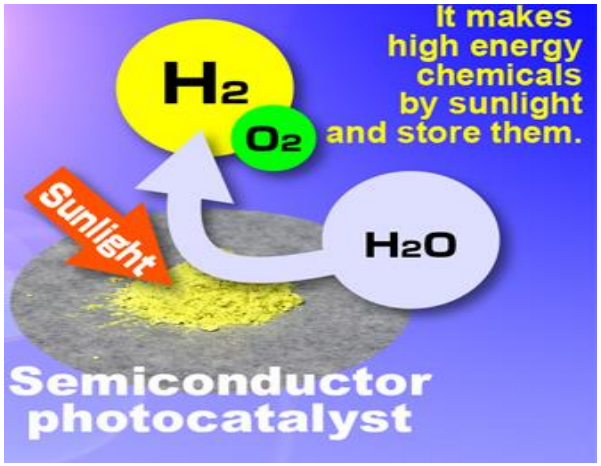
- Solar water heater
- Solar drying
- Solar Refigiration
- Solar cooking
- Solar distillation

Solar

to

Thermal





- Artificial Photosynthesis
- Solar water splitting

Solar

to

Chemical Fuels

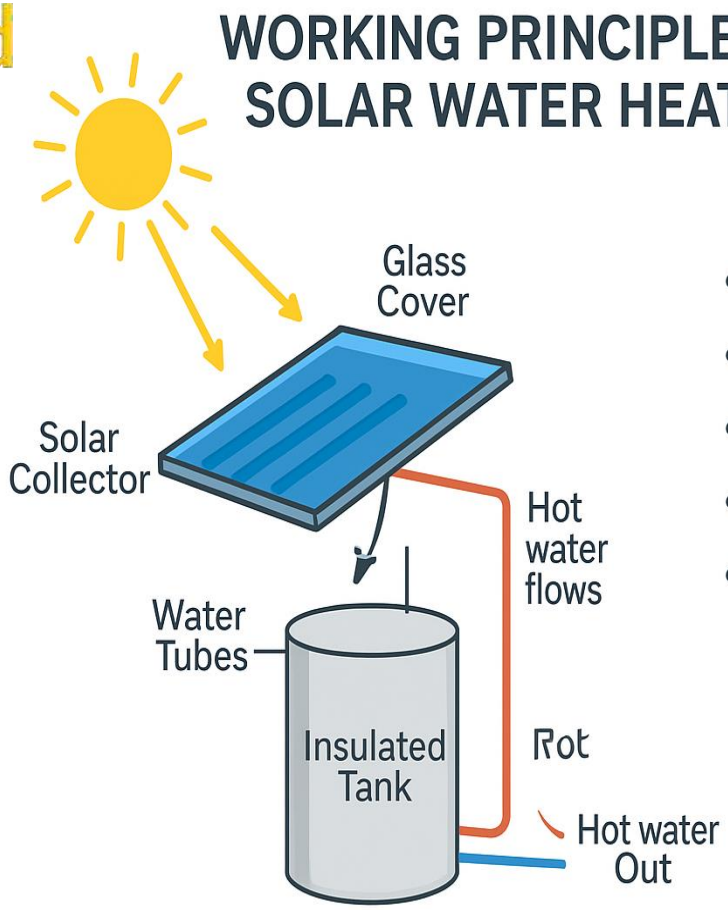
solar water heating system- types and mechanism

Solar water heater
Working principle

- The heat energy from the sun is observed by the panels and the process of conversion of heat energy into electric energy takes place through various process

Components used

- Solar panels
- Collectors
- controllers
- Pumps
- Tank as heat transfers and heat exchangers



- Absorption of Solar Energy
- Heating of Water
- Circulation of Water
- Storage of Hot Water
- Supplementary Heating (if required)

Sunlight absorption: Solar radiation (mainly visible and infrared light) falls on a surface or material.

Photon interaction: Photons from sunlight are absorbed by the material's surface (like black coatings, solar collectors, or water).

Conversion to heat: The absorbed **photons excite the material's atoms/molecules, increasing their vibrational energy** → this manifests as heat (thermal energy).

Heat transfer to medium: The generated heat is transferred to a working medium (e.g., water, air, or oil) through conduction, convection, or radiation.

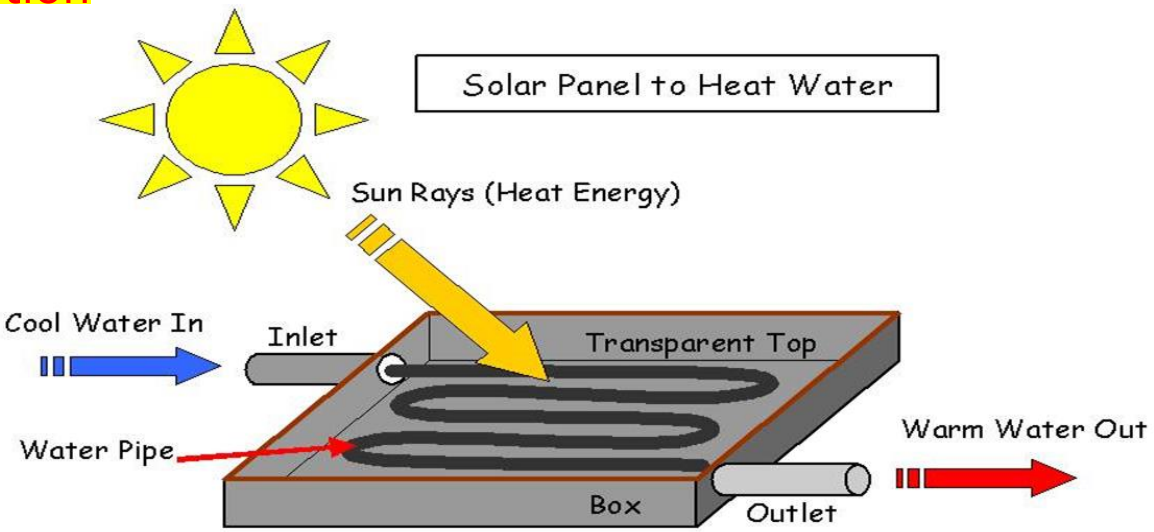
Temperature rise: The medium's temperature increases (e.g., water in a solar water heater becomes hot).

Storage/Utilization: Heat can be stored (e.g., in insulated tanks, molten salts, or phase change materials). Or directly used (for heating, cooking, drying, or power generation via steam turbines).

- Sunlight is **absorbed** by the solar collector.
- Absorber plate **converts solar radiation** into heat energy.
- Cold water flows through pipes/tubes in the **collector.**
- Water gets **heated** as it passes through.
- Hot water rises naturally (thermosiphon) or is **pumped to storage tank**.
- Insulated tank stores hot water for later use.

1: flat plate collectors

- In the flat plate collectors, the solar panel is covered with the glass in order to increase the heat observation



Flat plate collectors

•It can easily achieve a temperature 60-80°C above ambient temperature.

•It uses both beam and diffuse radiation.

•Does not require tracking.

•Requires little maintenance.

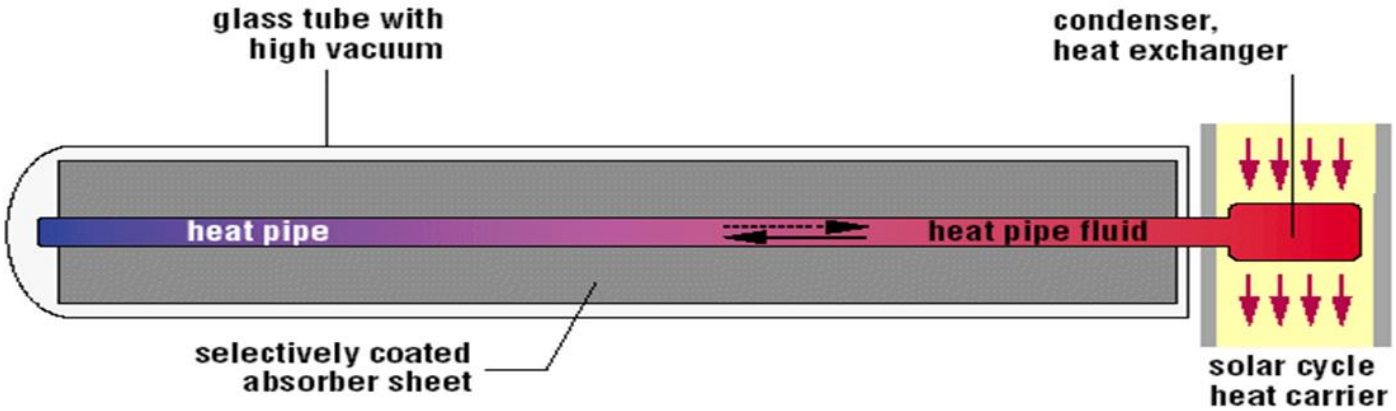
•Efficiency: ~ 45% at 80°C

•Applications: air heating, water heating, industrial process heating, passive air conditioning.



2.evacuated tube collectors

- This type has the evacuated double layer borosilicate glass tube. The inner tube is coated with the absorbing materials to efficiently absorb the sunlight. In this type the heat is lost to the atmosphere



•It can easily achieve a temperature 80-120°C above ambient temperature.

•It uses both beam and diffuse radiation.

•Does not require tracking.

•Requires little maintenance.

•Mode:Heat pipe;U tube water; Integrated Collector/storage

•Efficiency: ~ 55% at 80°C(for water heating)

•Applications: air heating, water heating, industrial process heating, passive air conditioning



Types of solar water heating

Active and passive are the two types of solar water heating system.

Active Solar Water Heating Systems

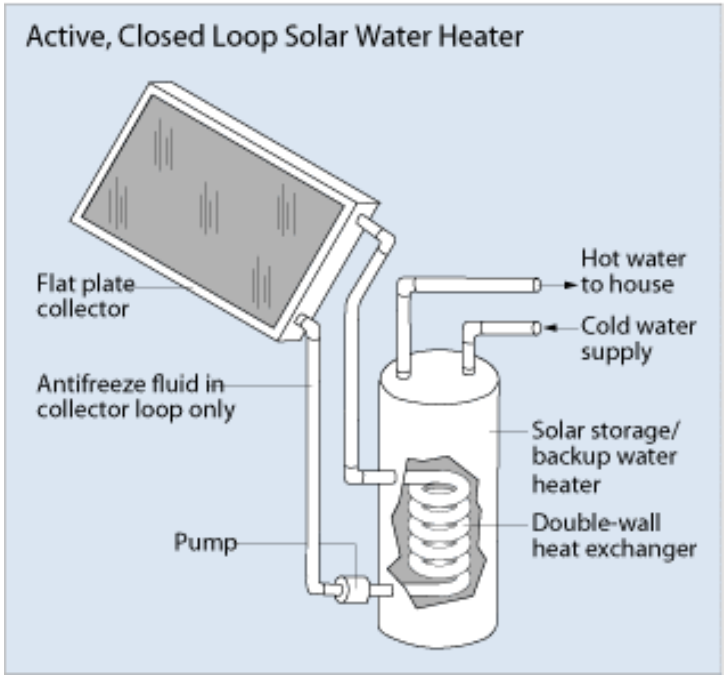
There are two types of active solar water heating systems:

- **Direct circulation systems**

Pumps circulate household water through the collectors and into the home. They work well in climates where it rarely freezes.

- **Indirect circulation systems**

Pumps circulate a non-freezing, **heat-transfer fluid** through the collectors and a **heat exchanger**. This heats the water that then flows into the home. They are popular in climates prone to freezing temperatures.



Passive Solar Water Heating Systems

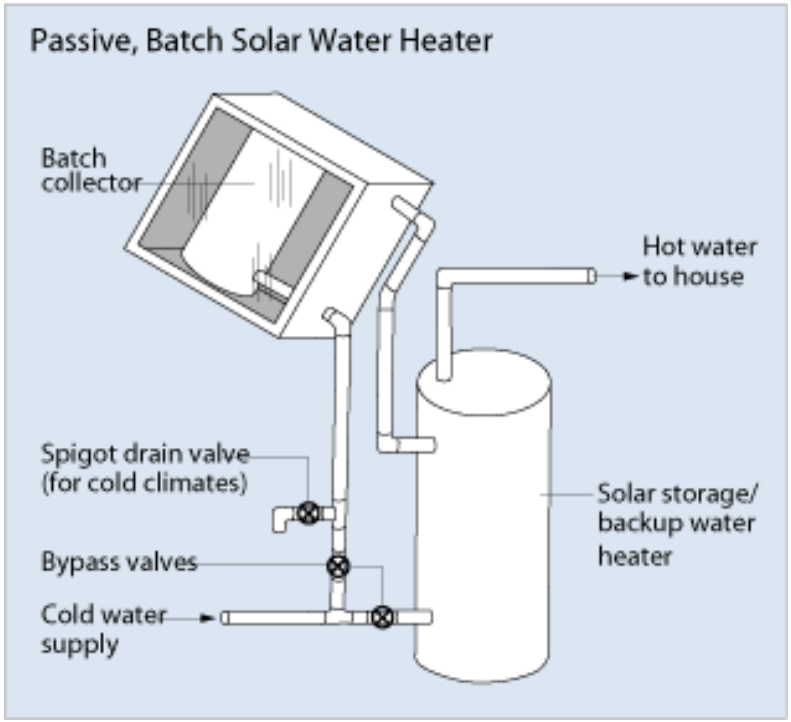
Passive solar water heating systems are typically less expensive than active systems, but they're usually not as efficient. However, passive systems can be more reliable and may last longer. There are two basic types of passive systems:

- **Integral collector-storage passive systems**

These consist of a storage tank covered with a transparent material to allow the sun to heat the water. Water from the tank then flows into the plumbing system. These work best in areas where temperatures rarely fall below freezing. They also work well in households with significant daytime and evening hot-water needs.

- **Thermosyphon systems**

Water is heated in a collector on the roof and then flows through the plumbing system when a hot water faucet is opened. The majority of these systems have a 40 gallon capacity.



Feature	Active System	Passive System
Working Principle	Uses pumps, controllers, and sometimes sensors to circulate water or heat-transfer fluid.	Relies on natural convection (thermosiphon) or gravity for water circulation, no pumps.
Complexity	More complex due to moving parts and electronic controls.	Simple design with fewer or no moving parts.
Efficiency	Generally more efficient, especially in colder climates, as circulation is controlled.	Less efficient; performance depends on natural circulation and sunlight availability.
Cost	Higher installation and maintenance cost.	Lower initial cost and minimal maintenance.
Maintenance	Requires periodic maintenance (pumps, sensors, controllers may fail).	Very low maintenance as no pumps or controllers are involved.
Reliability	Dependent on electrical components and pumps.	Highly reliable due to simplicity.
Installation	Flexible installation (can be installed in varied climates).	Works best in warm, sunny climates with good roof positioning.
Examples	Direct circulation system, Indirect circulation system.	Thermosiphon system, Integral collector storage system.

SOLAR REFRIGERATION

Refrigerator which runs on electricity provided by Solar Energy is known as solar refrigeration. Solar-powered refrigerators may be most commonly used in the future generation. Developing the world to help mitigate poverty and climate change.

It consists mainly two types:

Rechargeable:

It consist of a lead ion battery which stores charge for critical condition.

Non Rechargeable:

It runs directly on the solar panel.

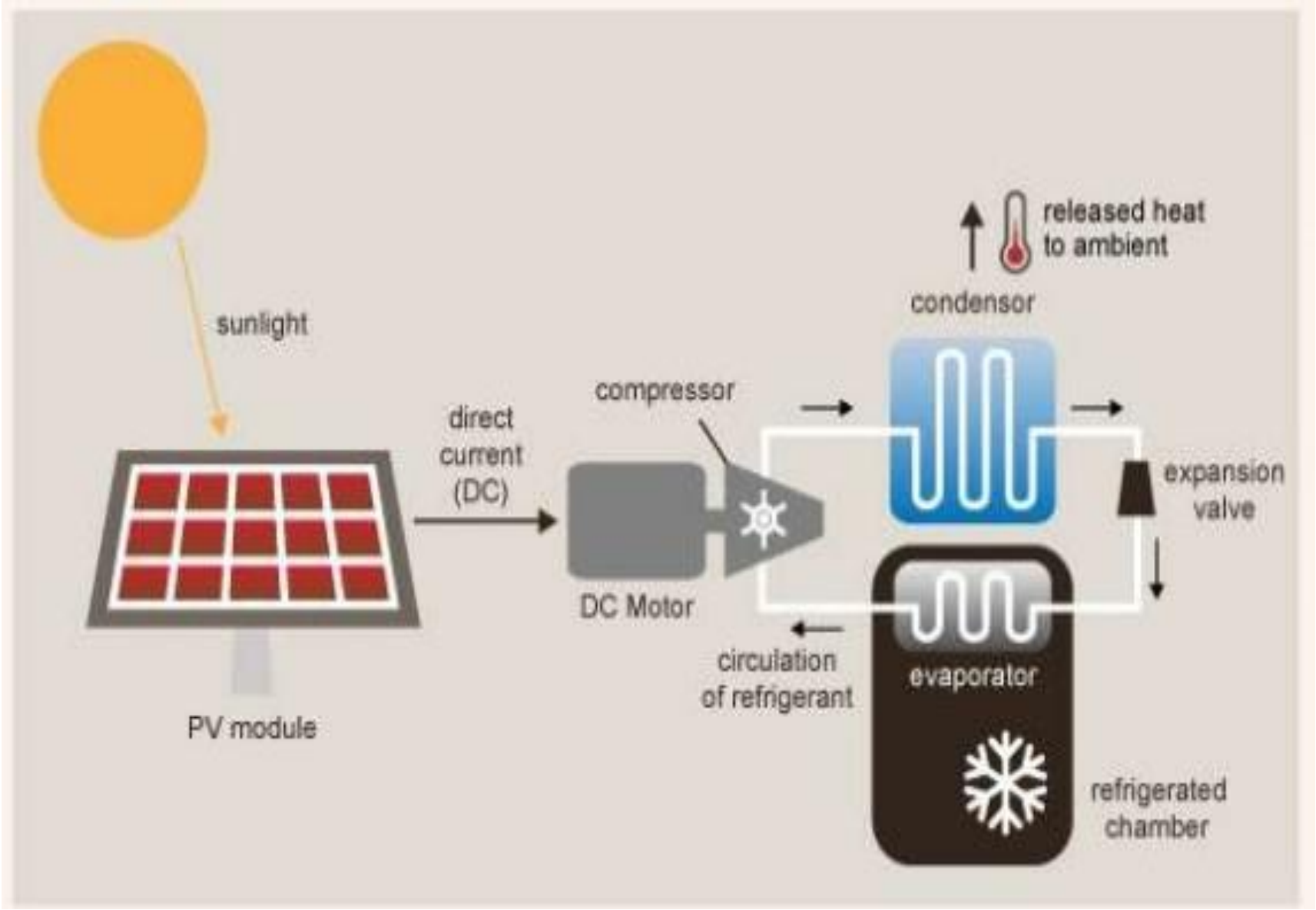
These are classified into three types:

a) Photovoltaic Operated Refrigeration Cycle

b) Solar Mechanical Refrigeration

c) Absorption Refrigeration

WORKING OF SOLAR REFRIGERATION SYSTEMS:



Working Principle

•Converts **solar energy into electricity** using **photovoltaic (PV) panels**.

•The generated electricity powers the **compressor** of a conventional vapor compression refrigeration cycle.

•The system operates like a normal refrigerator/air-conditioner, but instead of grid electricity, it uses **solar electricity**.

Mechanism (Step by Step)

1.Solar PV Panels → Convert sunlight into DC electricity.

2.Charge Controller & Battery (optional) → Store and regulate power supply for continuous operation.

3.Inverter (if AC compressor used) → Converts DC into AC electricity.

4.Compressor → Compresses refrigerant, increasing its pressure and temperature.

5.Condenser → Hot refrigerant releases heat to surroundings and condenses into liquid.

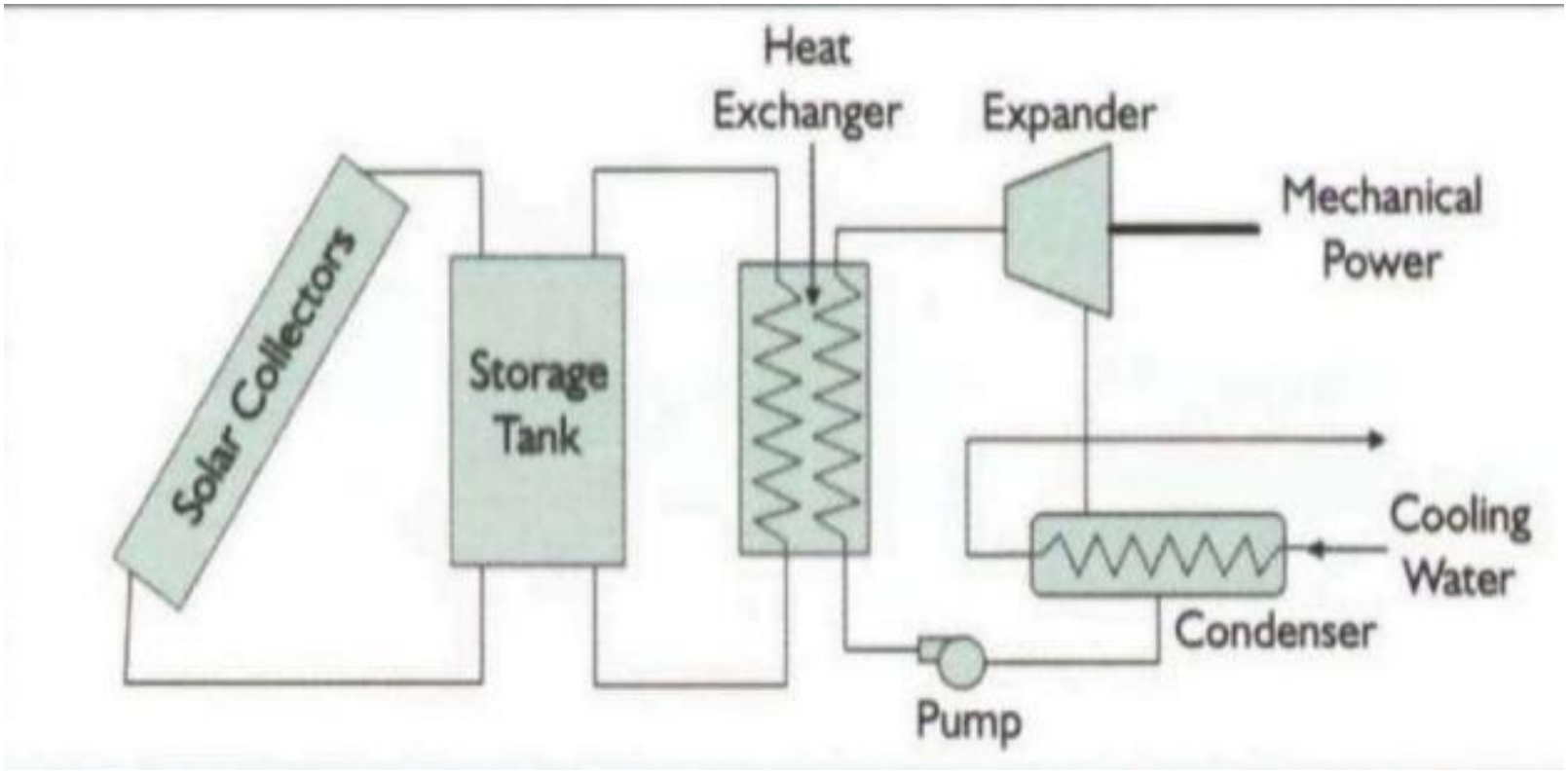
6.Expansion Valve → Liquid refrigerant pressure is reduced, cooling it.

7.Evaporator → Refrigerant absorbs heat from the refrigerated space, producing cooling.

8.Cycle repeats as long as solar power (or stored battery power) is available.

Solar Mechanical Refrigeration:

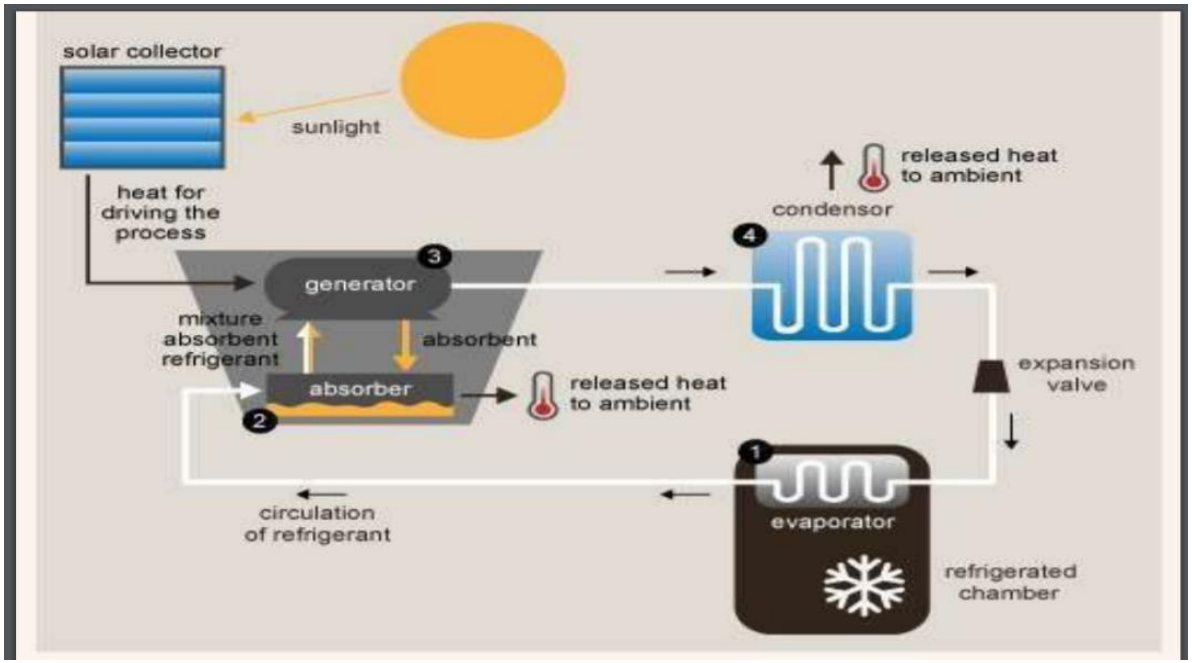
- Solar mechanical refrigeration uses a conventional vapour compression system driven by mechanical power that is produced with a solar-driven heat power cycle.
- The heat power cycle usually considered for this application is a Rankine cycle in which a fluid is vaporized at an elevated pressure by heat exchange with a fluid heated by solar collectors.
- A storage tank can be included to provide some high temperature thermal storage.
- The vapour flows through a turbine or piston expander to produce mechanical power.



- The fluid exiting the expander is condensed and pumped back to the boiler pressure where it is again vaporized.
- The efficiency of the Rankine cycle increases with increasing temperature of the vaporized fluid entering the expander.
- The Rankine cycle efficiency estimated for a high-temperature organic fluid assuming that saturated vapor is provided to a 70% efficient expander and condensation occurs at 35°C (95°F).
- The overall efficiency of solar mechanical refrigeration, defined as the ratio of mechanical energy produced to the incident solar radiation, is the product of the efficiencies of the solar collector and the power cycle.

Absorption Refrigeration System:

- Absorption refrigeration is the least intuitive of the solar refrigeration alternatives
- Unlike the PV and solar mechanical refrigeration options, the absorption refrigeration system is considered a "heat driven" system that requires minimal mechanical power for the compression process.
- It replaces the energy-intensive compression in a vapor compression system with a heat activated "thermal compression system."
- Ammonia is working fluid.
- Minimal mechanical power input (pump instead of compressor).
- Absorption into water solution allows it to be pumped.
- Desorbed in generator (rectifier required to separate out water).



Feature	**Absorption Refrigeration System**	**Solar Mechanical Refrigeration System**
Energy Source	Uses **heat energy** (from solar collectors, waste heat, or gas burners).	Uses **mechanical/electrical energy** (from PV panels or solar thermal engines).
Driving Mechanism	Uses a **thermal compressor** (generator + absorber + pump) instead of a mechanical compressor.	Uses a **mechanical/electrical compressor** driven by solar energy.
Working Fluid	Refrigerant + absorbent pair (e.g., Ammonia–Water, LiBr–Water).	Pure refrigerant (e.g., R134a, R22, R410a) in vapor compression cycle.
Efficiency (COP)	Generally **lower (0.6 – 0.8)**.	Relatively **higher (2.5 – 4.0)**.
Moving Parts	Very few moving parts (only small solution pump).	Has moving compressor, hence higher wear and tear.
Maintenance	Low maintenance (fewer mechanical components).	Requires more maintenance (compressor, motors, etc.).
Cost	Higher installation cost (due to large solar collectors, absorber setup).	Lower cost compared to absorption, especially with PV panels.
Suitability	Best for areas with **abundant solar heat** or waste heat (industrial use, remote cooling).	Best for areas with **strong solar radiation** and reliable PV systems (domestic, small-scale use).

- Heat into generator provided by solar collector.
- The pressurisation is achieved by dissolving the refrigerant in the absorbent, in the absorber section.
- Subsequently, the solution is pumped to a high pressure with an ordinary liquid pump.
- In this way the refrigerant vapour is compressed without the need of large amounts of mechanical energy that the vapour-compression air conditioning systems demand.
- This system greatly increases complexity.

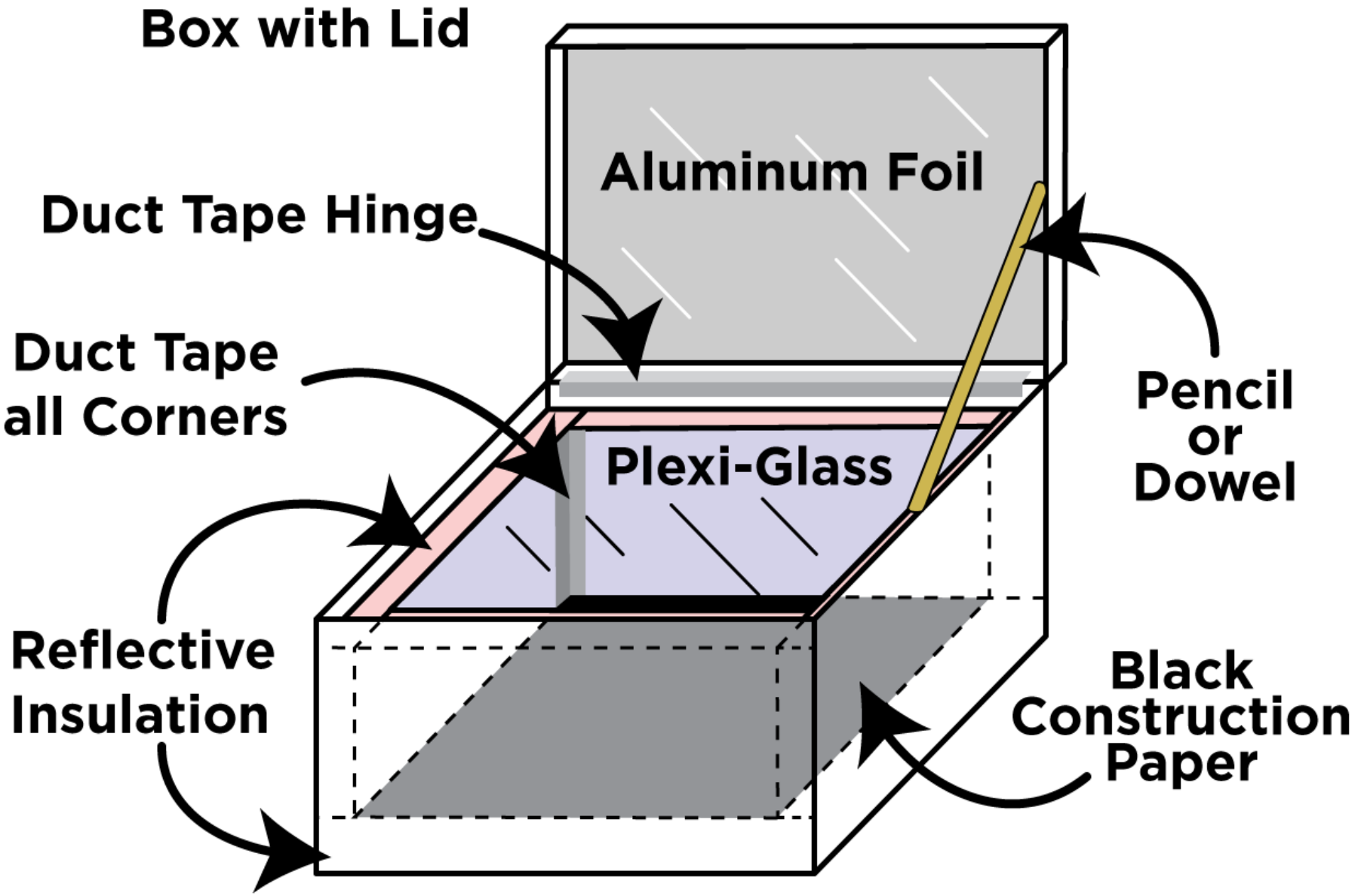
ADVANTAGES

- Solar energy is the main source of energy that is utilized to run solar refrigerator.
- So, significant amount of electrical power is saved and it also causes less pollution that would have been added due to the use of power produced by the conventional power plants.
- The solar energy is available in every part of the world and unlike fossil fuels and nuclear power, it is a clean source of energy. Additional power from the solar collector can also be used for the other domestic purposes.
- The solar refrigerators can be very useful where there is no continuous supply of electricity or difficult to get conventional fuel.
- More importantly it is renewable in nature. Conventional refrigeration systems emit significant amount of gas which pollute the environment.

DISADVANTAGES

- As solar radiation is not available throughout the day, power production is not uniform. Again it depends on the intensity of the beam radiation.
- Even in the hottest regions on earth, the average solar radiation flux rarely exceeds 1 kWh/m2 and the maximum radiation flux over a day is about 6 kWh/m2.
- These are low values from the point of view of technological utilization. So, those refrigeration systems can be used in those places where those problems are not present.
- To produce sufficient energy from solar system, it needs bigger collector.
- So, there is a need of bigger space for the collector which is another major problem for using solar refrigeration system.
- Initial investment to develop such set up is also large.

Solar Cooking



Solar Cooking

Key Components

1. Reflectors / Mirrors

To **concentrate or direct the sunlight into the cooking area**. Often shiny surfaces that reflect light.

2. Glazed Cover / Glass Lid

A transparent layer (glass or sometimes plexiglass) that allows sunlight in but reduces heat loss by trapping infrared radiation.

3. Blackened Absorbing Surface / Inner Box Painted Black

The cooking pot or inner surface is **dark or black to absorb as much light as possible**; this converts light to heat by conduction / radiation.

4. Insulation

To **reduce loss of heat to the surroundings**: sides, bottom (and occasionally top) are insulated so the heat remains inside.

5. Cooking Pot / Receiver

Where the food is placed. Often placed where the sunlight is focussed or inside the insulated chamber.

Working Principle (How It Works)

1.Collection of sunlight: Sunlight (visible + ultraviolet) reaches the cooker. Some designs use mirrors or reflectors to collect more sunlight (concentrating the solar energy).

2.Conversion to heat: The light is absorbed by black surfaces (pots, inner box) and converted to heat energy.

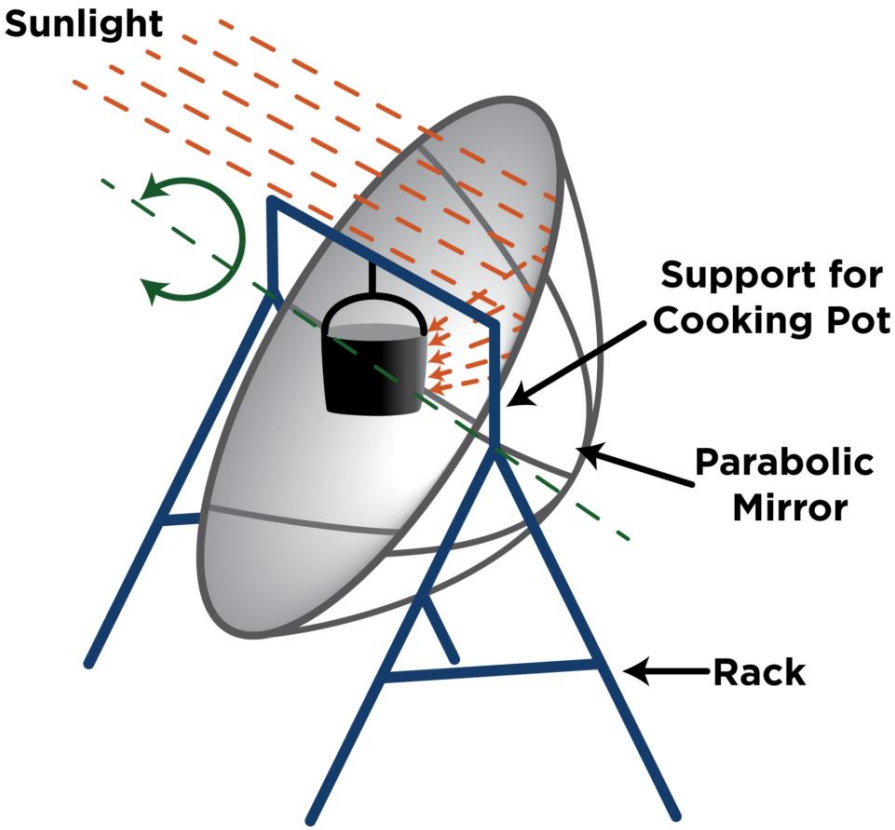
3.Trapping / Retaining heat: The glass lid (glazing) allows sunlight in, but traps heat inside since infrared radiation (heat) does not escape easily. Insulation and sealed chambers reduce convective and conductive heat losses.

4.Cooking process: The **temperature inside rises high enough to cook food**. For box cookers, temp may be lower but sufficient; for parabolic cookers, focal point can get quite hot.

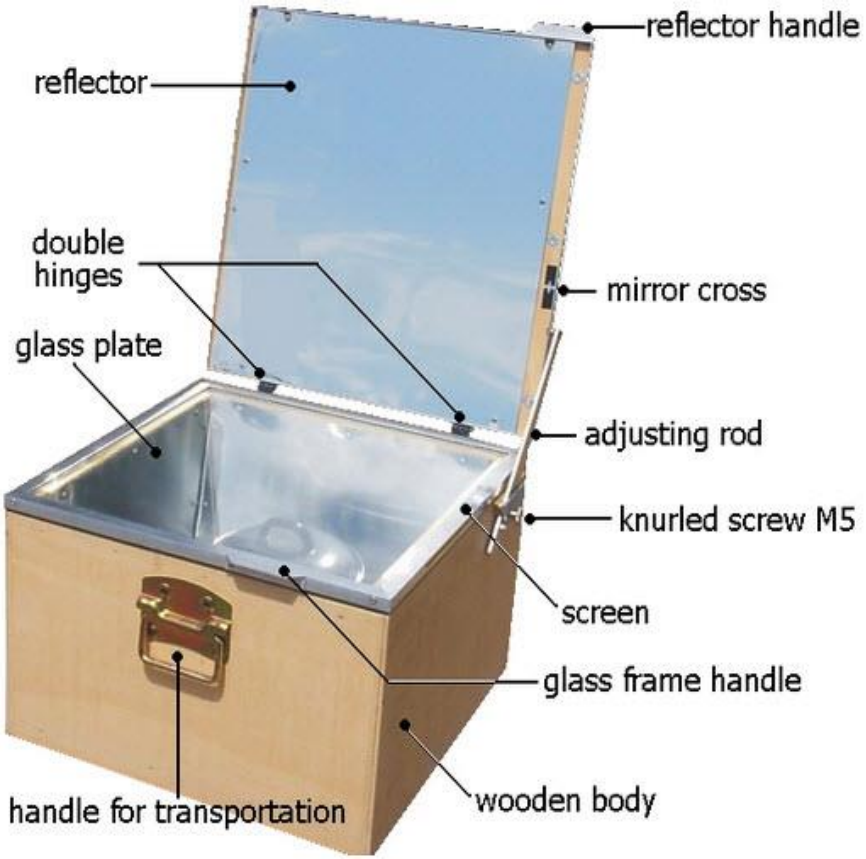
5.Adjustment with sun path: To **maintain efficiency,** reflectors / mirrors are adjusted as the sun moves, so that maximum sunlight is focused or entering the cooker.

Different Type of Solar Cooking

a. Parabolic solar cooker .

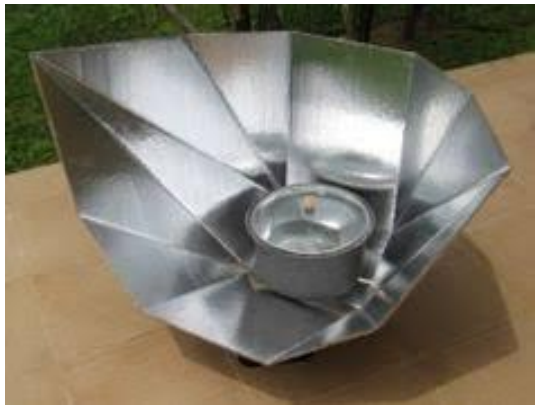


Solar box cooker

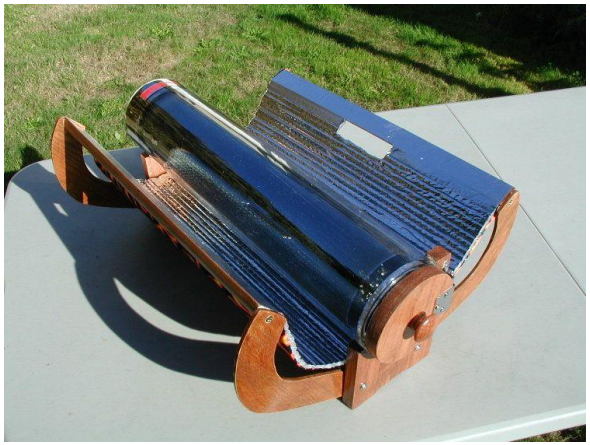


Different Type of Solar Cooking

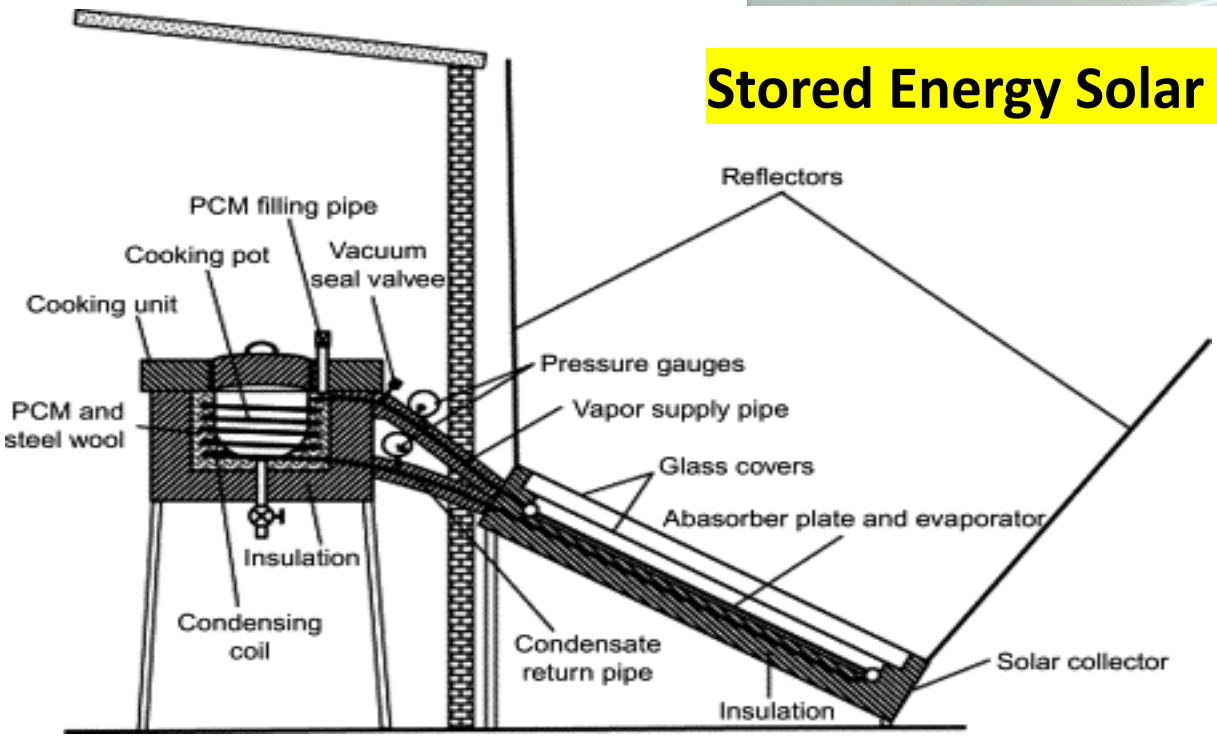
Panel Cooker



- Vaccum tube solar cooker



Stored Energy Solar cooker



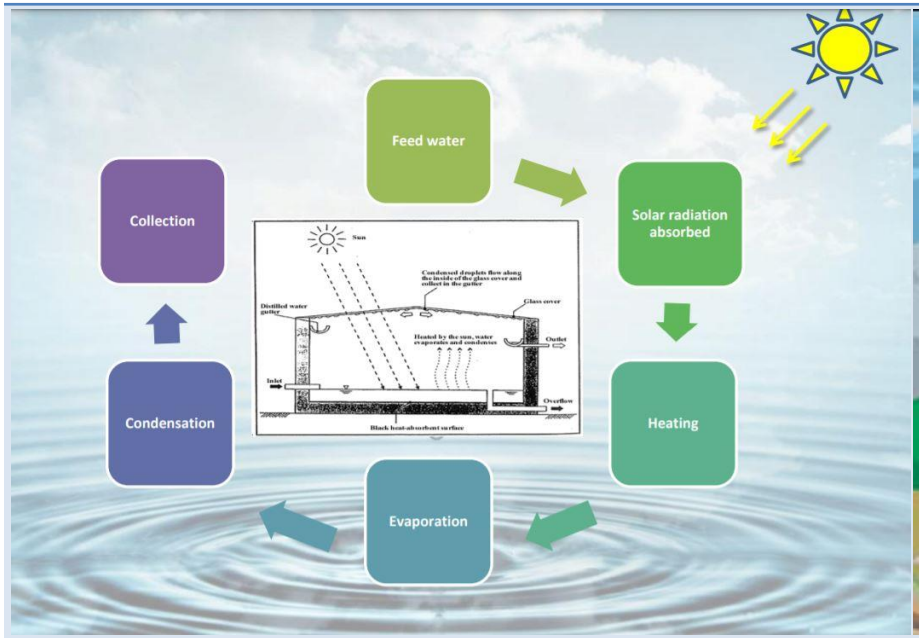
Introduction

Solar Distillation

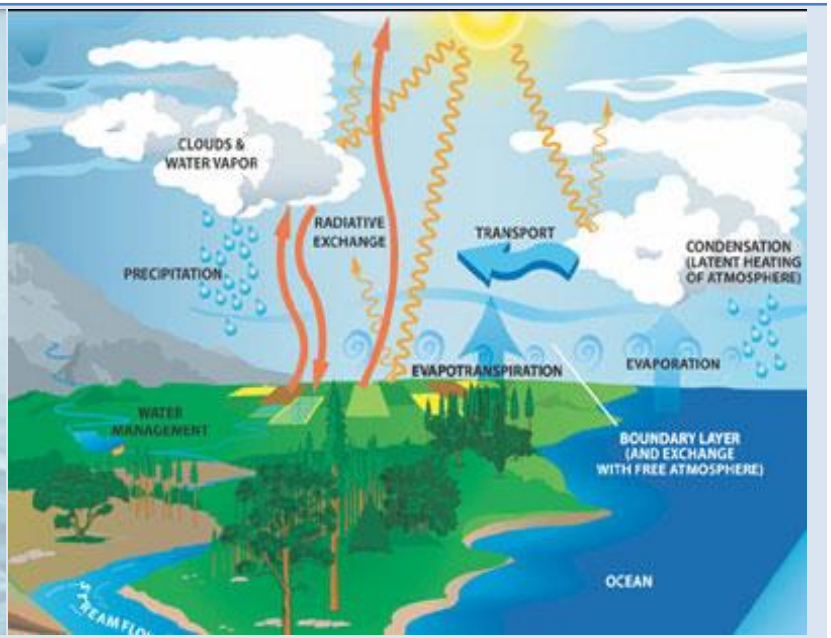
Solar distillation is one of the techniques in which low-grade solar energy is used for converting brackish/saline water into potable water.

Solar Distillation Analogy

Solar Distillation Process



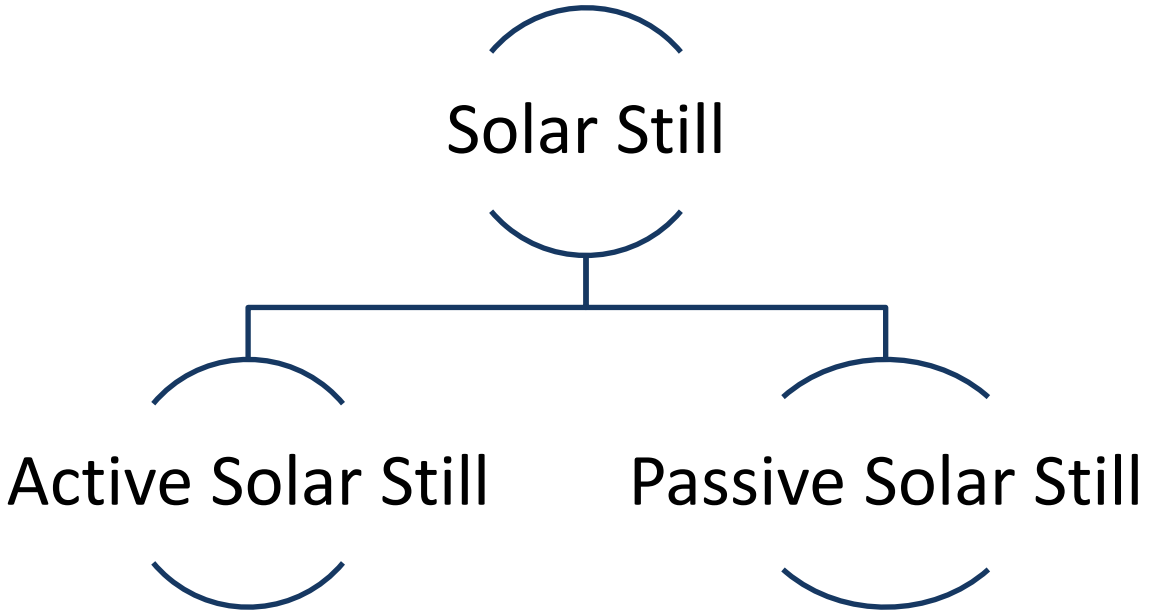
Natural Hydrologic Cycle



Energy Requirements of Water Distillation

- The latent heat of vaporization of water, is 2260 kilojoules per kilogram (kJ/kg).
- It should be noted that, although 2260kJ/kg is required to evaporate water, to pump a kg of water through 20m head, it requires only 0.2kJ/kg.
- Therefore we can say that solar distillation is mostly applicable at places where there is no local source of fresh water and can be easily pumped or lifted.

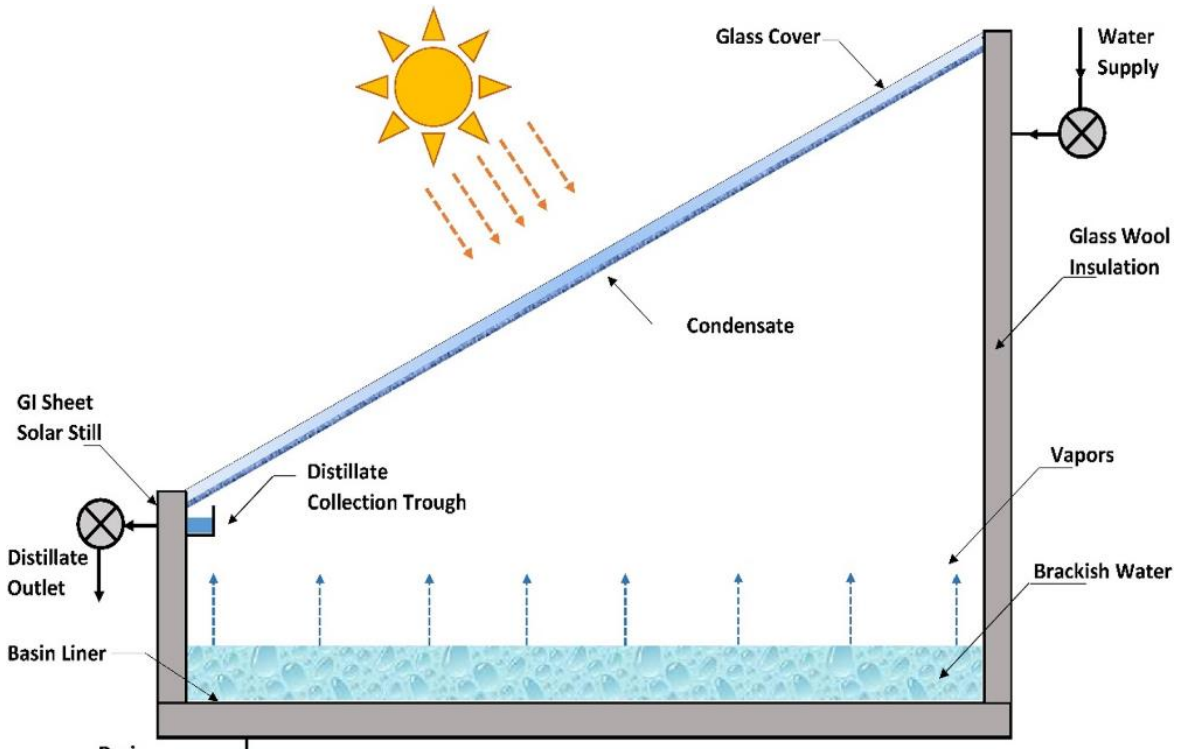
Solar Still- Device for Solar Distillation



An active solar still uses additional heat sources apart from sun.

A passive solar still is a conventional one that solely relies on the sun to distill water.

How a simple Solar Still works



- Sunlight enters **through the transparent** glass cover.
- Solar radiation heats the impure/salty water inside the basin.
- Water evaporates, leaving salts and impurities behind.
- The water vapor rises and condenses on the cooler glass cover.
- Condensed droplets trickle down into a collection channel.
- Collected water is pure distilled water.

Various types of Solar Stills

- Spherical Solar Still

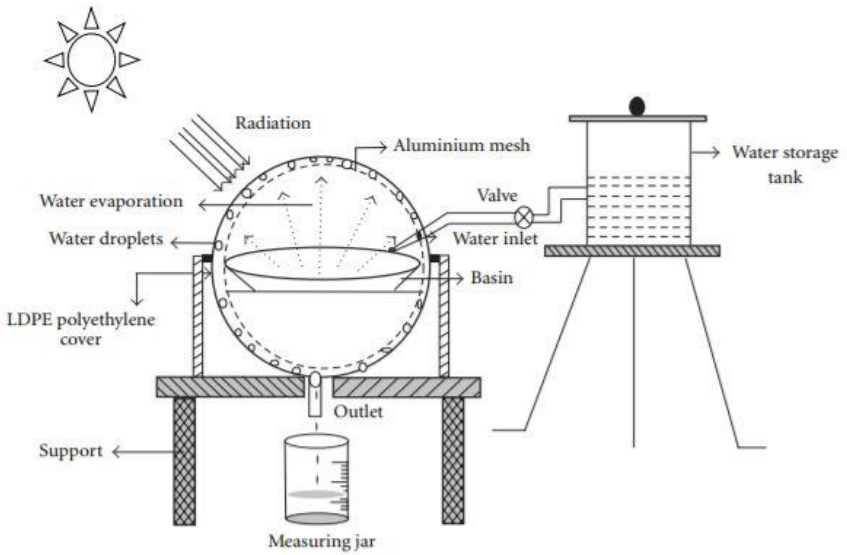


FIGURE 2: Schematic view of spherical solar still.



FIGURE 3: Pictorial view of Spherical Solar Still.

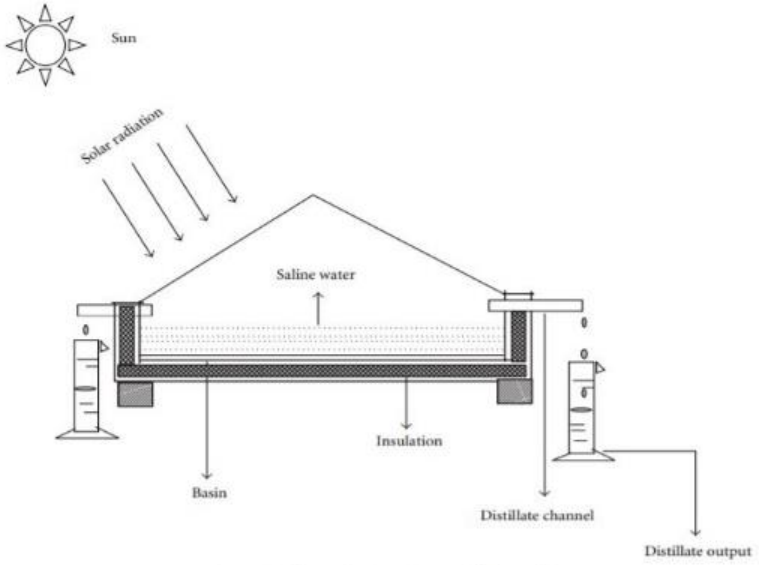


FIGURE 4: Schematic view of pyramid solar still.



FIGURE 5: Pictorial view of pyramid solar still.

- Hemispherical Solar Still

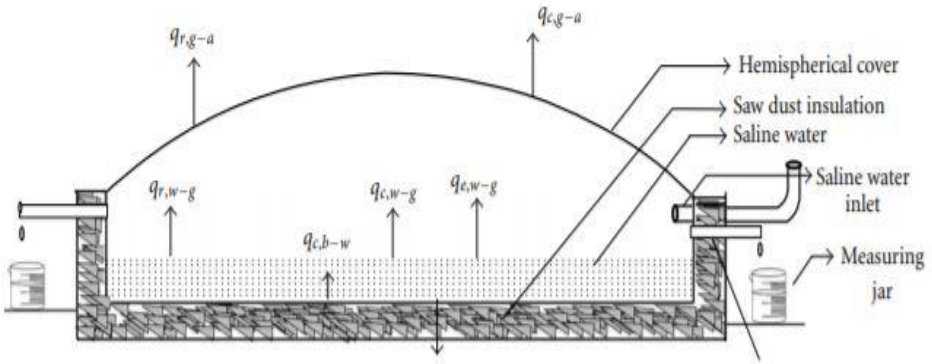


FIGURE 6: Schematic view of hemispherical solar still [8].



FIGURE 7: Pictorial view of hemispherical solar still.

- Double Basin Solar Still

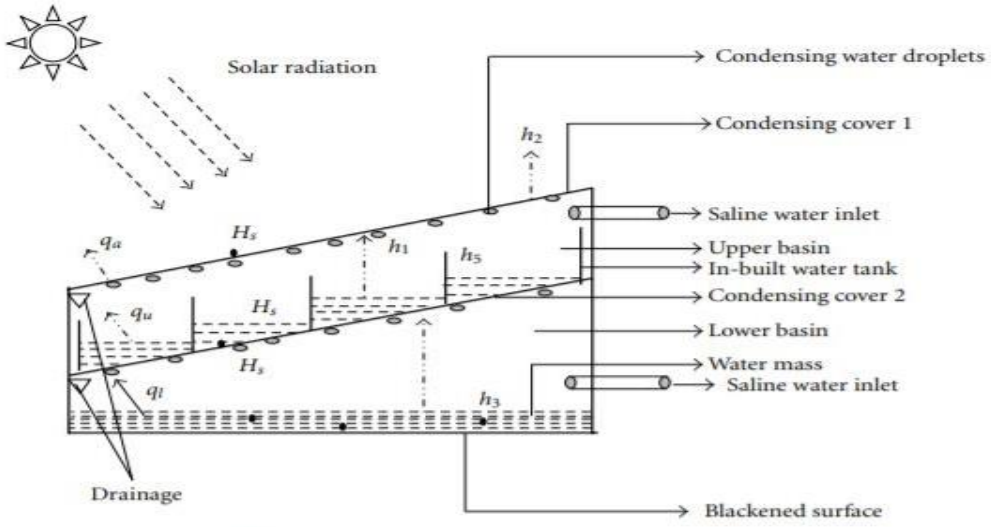


FIGURE 8: The cross-sectional view of double-basin glass solar still.

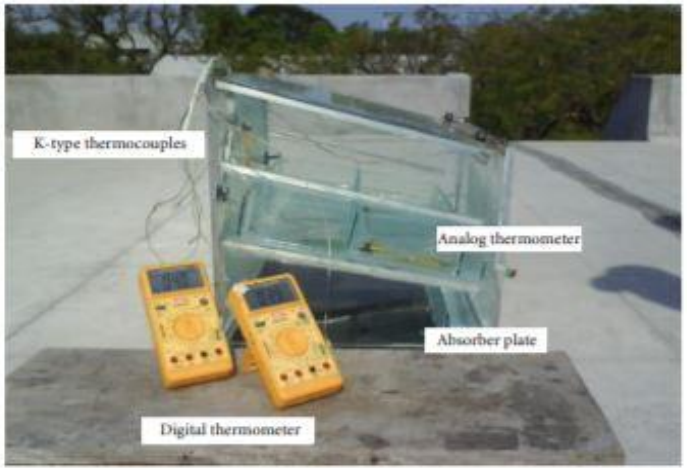


FIGURE 10: Photographic view of the double-basin glass solar still (side view).

- Tubular Solar Still
- TSS Pyramidal Solar Still

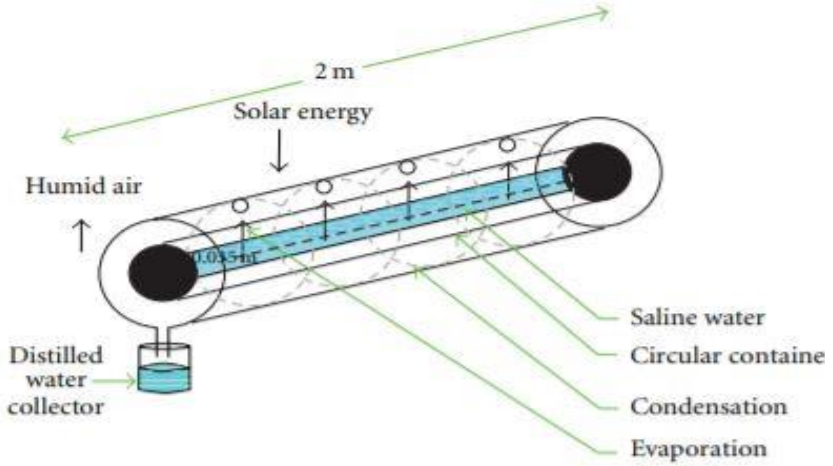


FIGURE 11: Schematic view of tubular solar still.

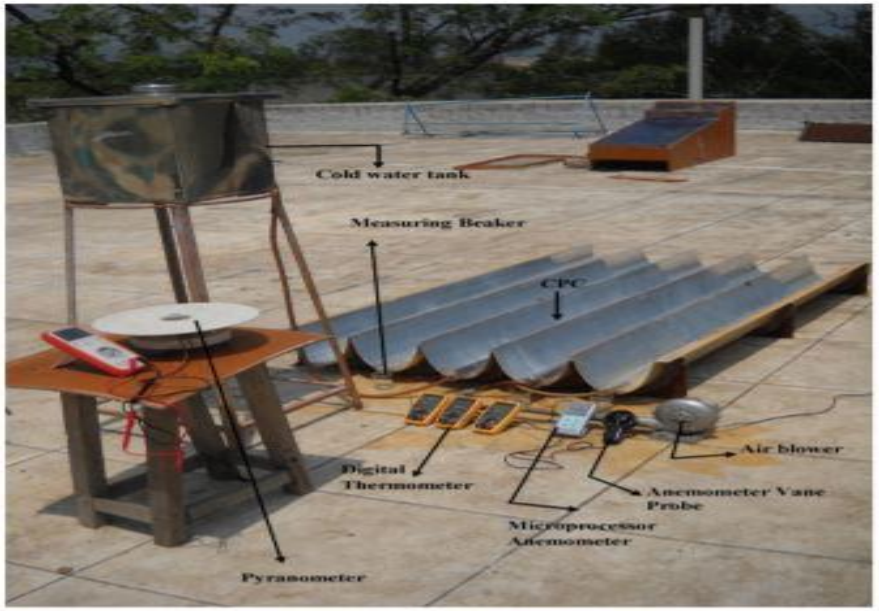


FIGURE 12: Pictorial view of CPC-TSS.

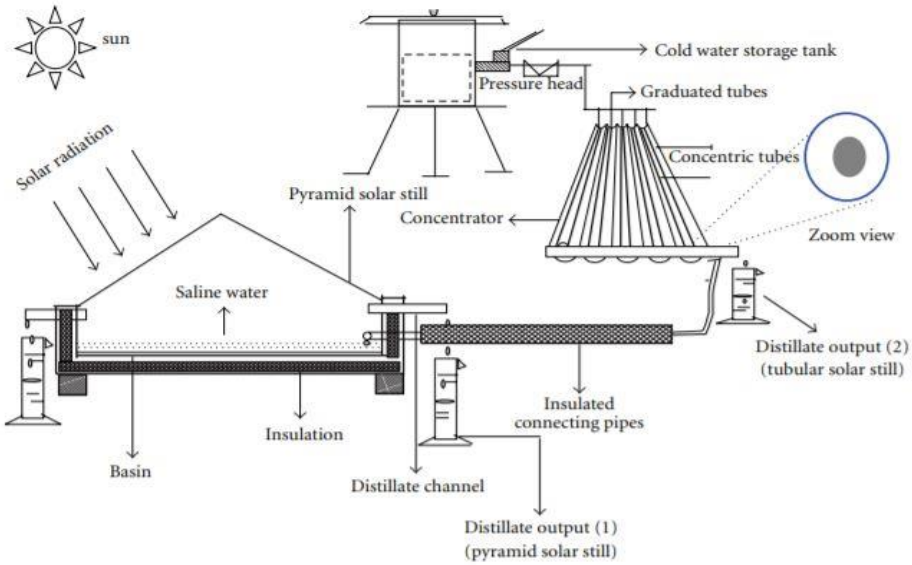
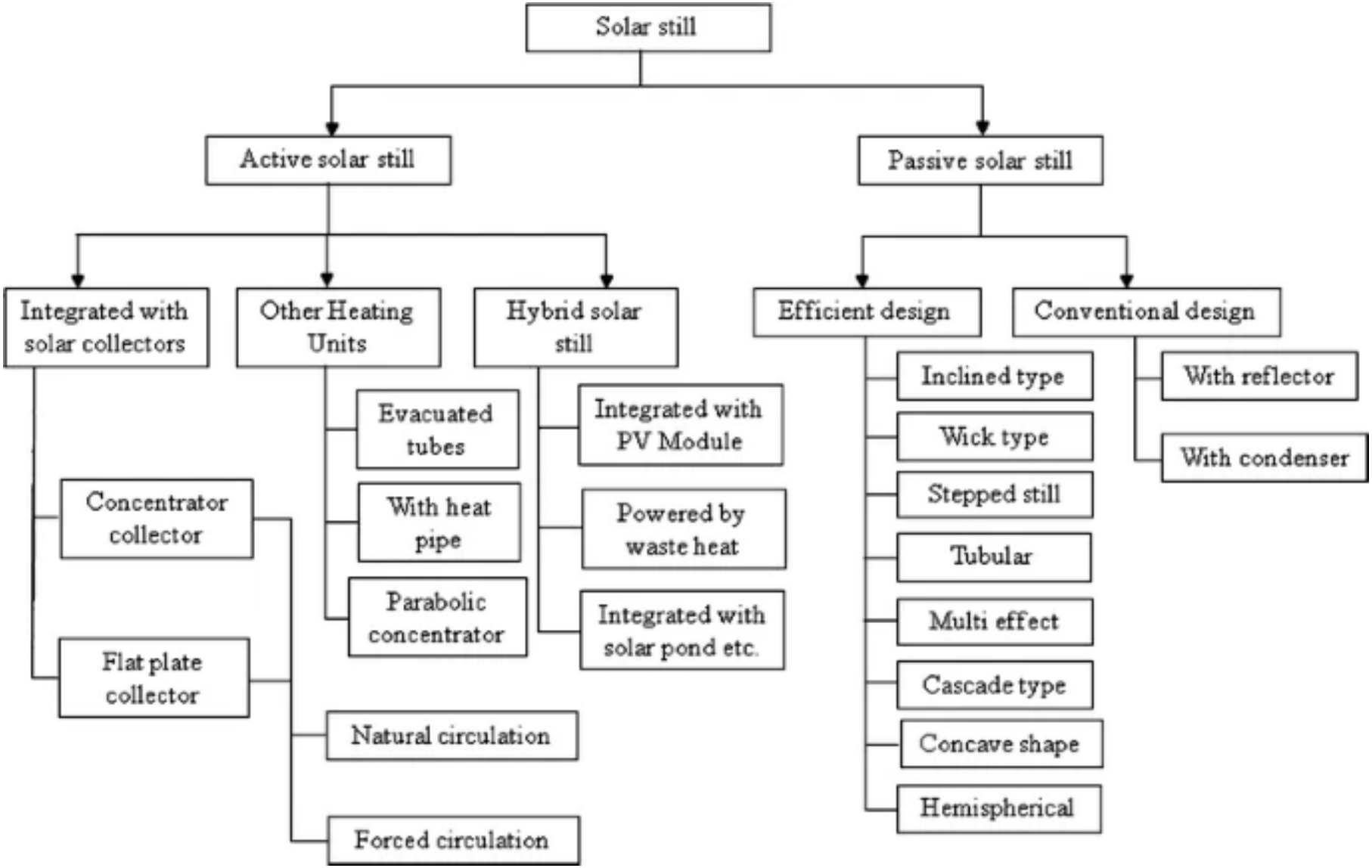


FIGURE 15: Cross sectional view of tubular solar still coupled with pyramid solar still.



FIGURE 16: Pictorial view of CPC-pyramid solar still design.

Brief Classification of types of Solar Stills



Design Objectives for an Efficient Solar Still

For high efficiency the solar still should maintain :

- A high feed (undistilled) water temperature
- A large temperature difference between feed water and condensing surface
- Low vapor leakage.

A high feed water temperature can be achieved if :

- Low absorption glazing and a good radiation absorbing surface are required
- Heat losses from the floor and walls are kept low
- The water is shallow so there is not so much to heat.

A large temperature difference can be achieved if :

- The condensing surface absorbs little or none of the incoming radiation
- Condensing water dissipates heat which must be removed rapidly from the condensing surface .

BENEFITS

Environmental Benefits

- Diversifies the sources of freshwater and therefore reduces stress on local freshwater reservoirs.
- Uses a renewable and free source of energy.

Socioeconomic Benefits

- Bears no associated operational energy costs and installation is relatively cheap.
- Provides safe drinking water from saltwater or soft, alleviating water stress and health risks from contaminated drinking water.

Pros. and Cons.

Advantages	Disadvantages
Solar distillation units are easy to construct, simple to operate and low maintenance	Rate of distillation is usually very slow
Solar distillation produces 99.9% pure water for consumption	The system can only be used when there is a supply of thermal energy readily available
Units do not have a carbon footprint as they are 100% clean and emit absolutely no harmful toxins into the atmosphere	Decreased productivity with time
Solar energy is renewable, abundant, has a widespread distribution	The system will not be efficient in countries with little or no solar energy available
Solar water distillers can be used at the household level, as well as scaled up through programmatic approaches	Requires some skills and time to maintain

Solar Drying

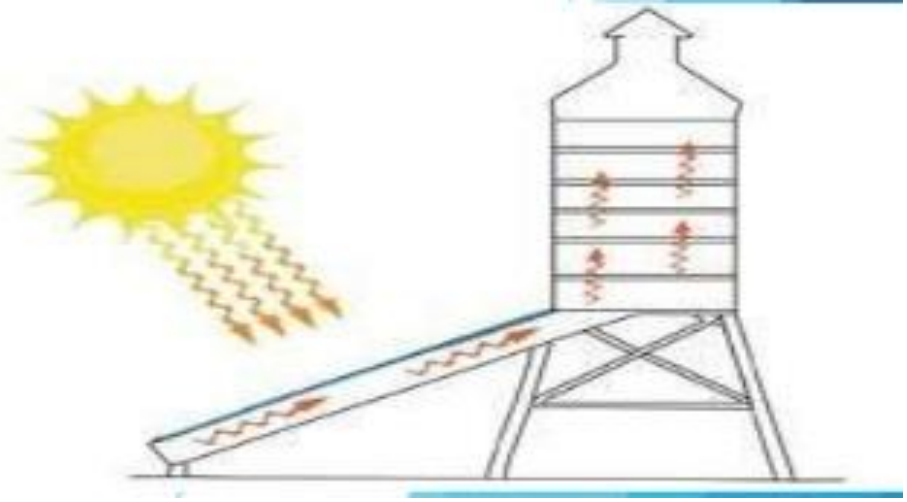
Introduction

Drying

Drying is a heat and mass transfer process consisting of the removal of moisture by evaporation from a solid , semi-solid or liquid.

- **Main objectives:-**

1. Extended storage life
2. Quality enhancement
3. Ease of handling
4. Further processing



Modes of Heat Transfer

Radiation

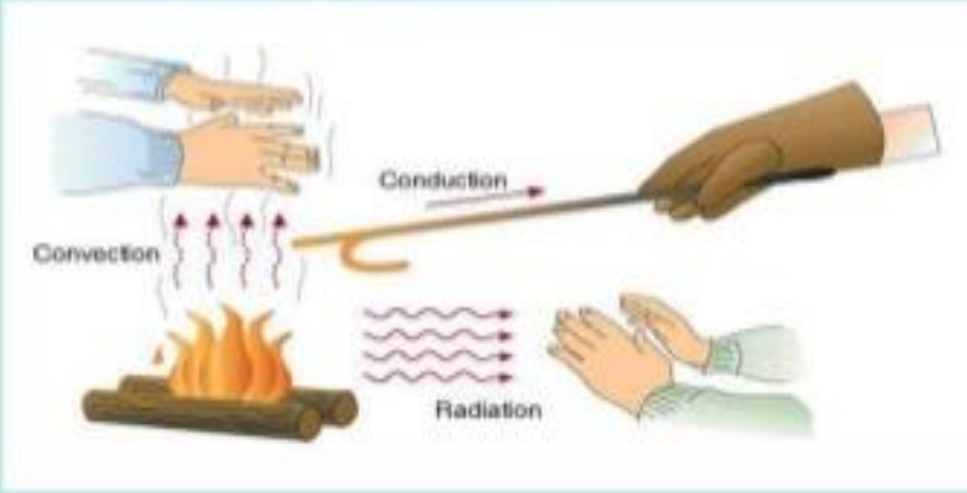
- Heat travels across space and in the atmosphere

Conduction

- Heat from a warmer object flowing to a colder object when in direct contact

Convection

- Heat transfer by currents (Hot air rises, cold air sinks



Can be used for drying of:-

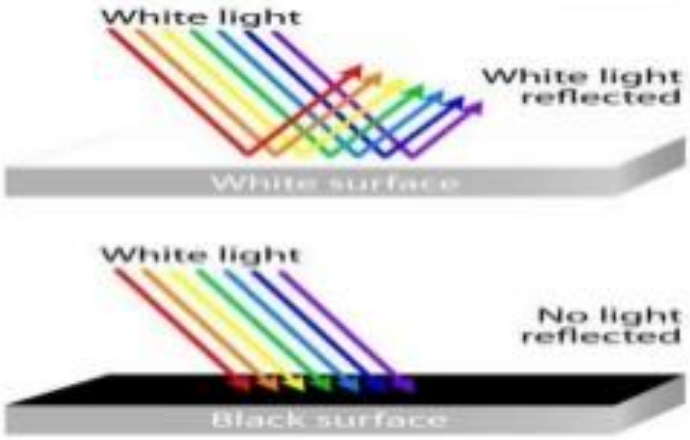
- Fruits
- Vegetables
- Fish
- Meat
- Mushrooms
- Medical plant
- Spices
- Tea

Working Principle of Solar Dryer

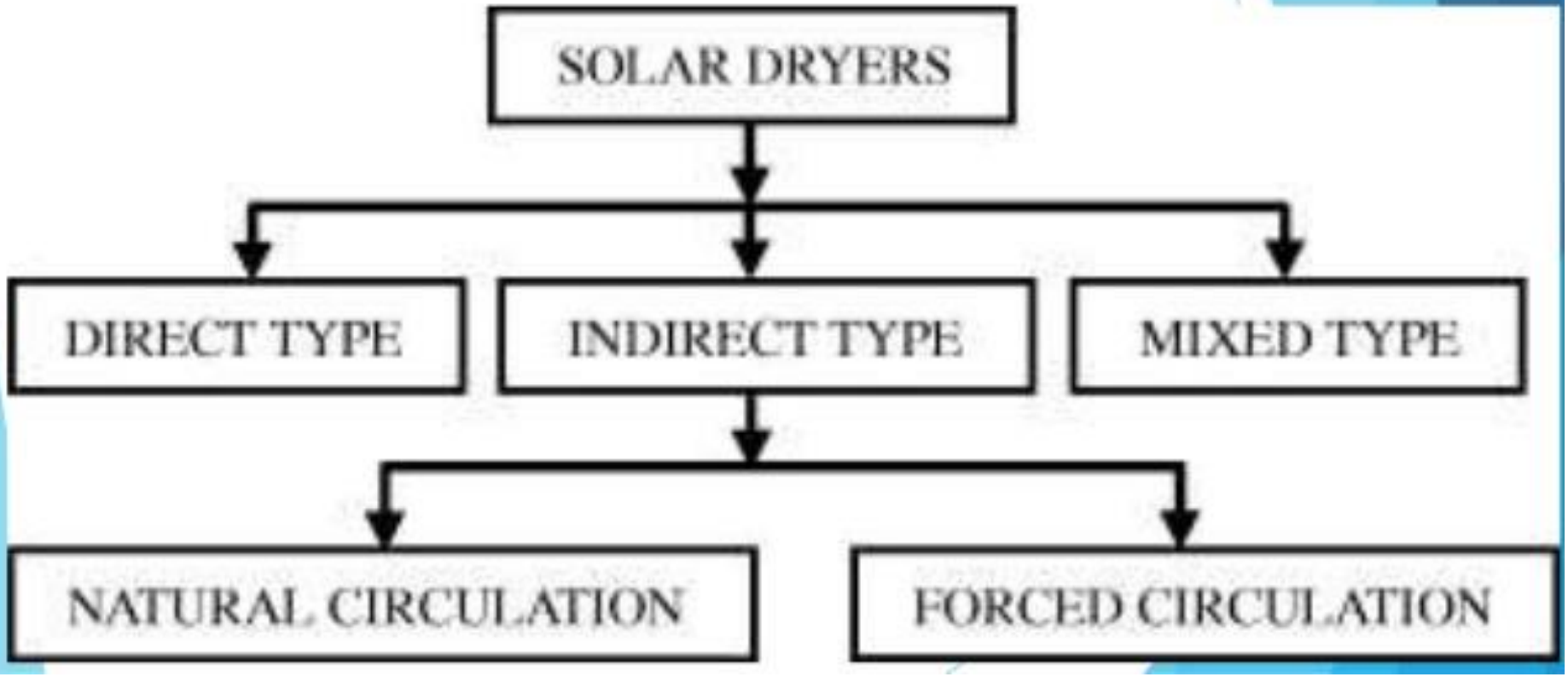
- **Conservation of energy** in which one form of energy is converted into other form of energy



- **Black body** which absorbs the heat of the incident sunlight and makes the temperature of chamber more than the surrounding temperature.

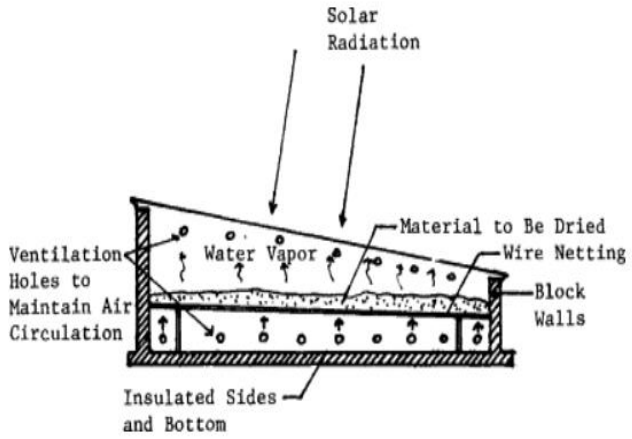


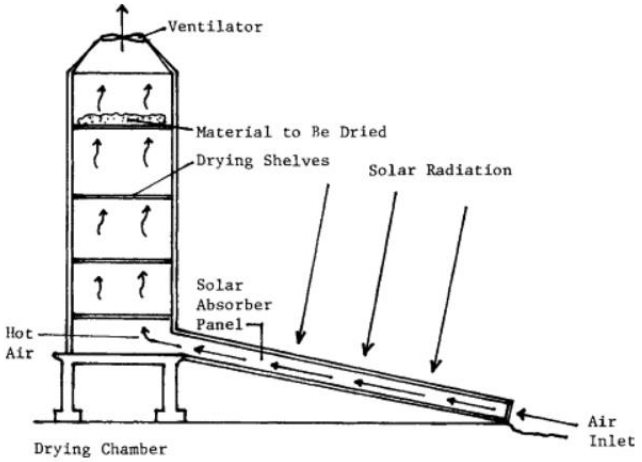
Classification of Solar Dryers



Passive

Active





Active vs Passive (based on air circulation)

Solar Dryers the most important characteristics to be considered are:

- Low capital and operation cost.
- High load ratio.
- Use of commercially available technology.
- Achievement of uniform material drying.
- Have a high throughput of fruit or other crops.
- Have simple mechanical material handling system.

Compared with open-air drying

- The product is hygienic.
- Protection of rain.
- Protection of pollution by dust etc.
- Protection of wind which can blow away the food.
- Reduction of drying time.
- No loss of colours.

Advantages

- **Efficient Heating Device:** These dryers work in any sunny area which include decks, patios, gardens and so forth.
- **Portable:** Most of these devices are compact in size and extremely portable. It is also light in weight which is one of the greatest advantages of using these devices.
- **Operational Costs:** It also involves zero operational costs and it can function on passive solar energy too.
- **Temperature Control:** It is also quite easy to control the temperature as it includes adjustable venting features.
- **Weatherproof Designs:** These devices also include sturdy and weather proof designs which ensure long time usage and durability.
- **Direct And Indirect Heating:** These devices utilize both direct and indirect solar energy for quick drying purposes. It also offers optimum performance and so the food items last for a longer period of time.

Disadvantages

- Not workable at night.
- Efficiency decreases to a large extent on cloudy days.
- Over heating may occur if regular attention is not paid.
- Change in taste and flavour of food may occur if regular monitoring is not done.
- We can not get the accurate amount of sun rays and heat.